Phil Dyke-Introduction to Laplace Transforms and Fourier Series Notes

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August 2026

Abstract

This is the digitization of the notes and exercises to Phil Dyke's book on Laplace Transforms and Fourier series from Springer Undergraduate Mathematics Series. This is the first math book that I've read cover to cover and I quite like it. It was part of the reason why I switched into applied math and provides a lot of useful insights that help with higher level math. Part of the later exercises may not be included due to work done on a chalkboard (and I was too lazy to copy them over) or in another notebook which was lost due to water damage unfortunately.

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Chapter 1

Introduction

1.1 Elementary Properties

For a function $F(x)$ to have a Laplace transform, it must be of exponential order and be piecewise continuous on the interval $[0, \infty)$. To be of "exponential order" means that there exists an $M, \alpha \in \mathbb{R}$ and $M, \alpha > 0$ such that $|F(x)| \leq Me^{\alpha t}$. We also assume that the function is Riemann integrable. We shall begin by proving some basic properties. Phil Dyke adopts the convention

$$\mathscr{L}\{F(t)\} = f(s)$$

and as such, I will be adopting the same convention for consistency.

Theorem 1.1 (Linearity)

$$\mathscr{L}\{aF_1(t) + bF_2(t)\} = a\mathscr{L}\{F_1(t)\} + b\mathscr{L}\{F_2(t)\}$$

Proof.

$$\begin{aligned}\mathscr{L}\{aF_1(t) + bF_2(t)\} &= \int_0^\infty (aF_1 + bF_2)e^{-st}dt \\ &= a\int_0^\infty F_1 e^{-st}dt + b\int_0^\infty F_2 e^{-st}dt \\ &= a\mathscr{L}\{F_1(t)\} + b\mathscr{L}\{F_2(t)\}\end{aligned}$$

This is assuming both F_1 and F_2 are of exponential order such that

$$|F_1(t)| \leq M_1 e^{\alpha_1 t} \qquad |F_2(t)| \leq M_2 e^{\alpha_2 t}$$

and

$$\begin{aligned}|aF_2 + bF_2| &\leq |a||F_1| + |b||F_2| \\ &\leq (|a|M_1 + |b|M_2)e^{\max\{\alpha_1, \alpha_2\}t}\end{aligned}$$

□

Theorem 1.2 (First Shift Theorem) *If $F(t)$ is of exponential order such that $|F(t)| \leq Me^{\alpha t}$ for two positive numbers $M, \alpha \in \mathbb{R}$, then $\mathscr{L}\left\{e^{-bt}F(t)\right\} = f(s+b)$ if $b \leq \alpha$.*

Proof.

$$\begin{aligned}\mathscr{L}\left\{e^{-bt}F(t)\right\} &= \int_0^\infty e^{-bt}F(t)e^{-st}dt \\ &= \int_0^\infty F(t)e^{-(s+b)t}dt \\ &= f(s+b)\end{aligned}$$

□

We shall now find the Laplace transform of $F(t) = t$. We shall use integration by parts.

$$\begin{aligned}\mathscr{L}\{t\} &= \int_0^\infty te^{-st}dt \\ &= -\frac{t}{s}e^{-st}\Big|_0^\infty - \frac{1}{s^2}e^{-st}\Big|_0^\infty \\ &= 0 - \frac{1}{s^2}(0-1) \\ &= \frac{1}{s^2}\end{aligned}$$

Corollary 1.3 *The Laplace transform of t^n is*

$$\mathscr{L}\{t^n\} = \frac{n!}{s^{n+1}} \tag{1.1}$$

Proof.

$$\begin{aligned}\mathscr{L}\{t^n\} &= \int_0^\infty t^n e^{-st}dt \\ &= -\frac{t^n}{s}e^{-st}\Big|_0^\infty + \int_0^\infty \frac{nt^{n-1}}{s}e^{-st}dt \\ &= \frac{n}{s}\mathscr{L}\left\{t^{n-1}\right\}\end{aligned}$$

One can verify the this result easily for the base case $n = 2$. Now let's establish the recurrence relation

$$\mathscr{L}\left\{t^{n+1}\right\} = \frac{n+1}{s}\mathscr{L}\{t^n\} = \frac{n+1}{s}\frac{n!}{s^{n+1}} = \frac{(n+1)!}{s^{n+2}}$$

□

The only trigonometric functions with valid Laplace Transforms are cosine and sine as the others all have singularities in finite time. We shall now find the Laplace Transforms of these two functions by considering the Laplace transform of e^{it} by recalling Euler's formula.

$$\begin{aligned}e^{it} &= \cos t + i\sin t \\ \mathscr{L}\left\{e^{it}\right\} &= \mathscr{L}\{\cos t\} + i\mathscr{L}\{\sin t\}\end{aligned}$$

We get the above relation by the linearity property of the Laplace Transform and therefore the real part and imaginary parts of $\mathscr{L}\left\{e^{it}\right\}$ will correspond to the transforms of cos and sin respectively.

$$\begin{aligned}\mathscr{L}\left\{e^{it}\right\} &= \int_0^\infty e^{(i-s)t}dt \\ &= \frac{e^{(i-s)t}}{i-s}\Big|_0^\infty \\ &= \frac{1}{s-i}\frac{s+i}{s+i} = \frac{s}{s^2+1} + i\frac{1}{s^2+1}\end{aligned}$$

We have therefore established that

$$\mathscr{L}\{\cos t\} = \frac{s}{s^2+1} \qquad \mathscr{L}\{\sin t\} = \frac{1}{s^2+1}$$

If we do the same thing but for a generalized exponential e^{ait} we get that

$$\mathscr{L}\left\{e^{ait}\right\} = \frac{s}{s^2+a^2} + i\frac{a}{s^2+a^2}$$

Example 1.4 Find the Laplace transform of the sawtooth function

$$F(t) = \begin{cases} t & 0 \le t \le t_0 \\ 2t_0 - t & t_0 \le t \le 2t_0 \\ 0 & t \ge 2t_0 \end{cases}$$

$$\begin{aligned} \mathscr{L}\{F(t)\} &= \int_0^{t_0} te^{-st}dt + \int_{t_0}^{2t_0} (2t_0 - t)e^{-st}dt \\ &= \frac{4}{s^2} e^{-st_0} \sinh^2\left(\tfrac{1}{2}st_0\right) \end{aligned}$$

Theorem 1.5 *If* $\mathscr{L}\{F(t)\} = f(s)$ *then* $\mathscr{L}\{tF(t)\} = -\frac{d}{ds}f(s)$ *and generally* $\mathscr{L}\{t^nF(t)\} = (-1)^n \frac{d^n}{ds^n}f(s)$

Proof.

$$\frac{d}{ds}\mathscr{L}\{F(t)\} = \frac{d}{ds}\int_0^\infty F(t)e^{-st}dt = -\int_0^\infty te^{-st}dt = -\mathscr{L}\left\{te^{-st}\right\}$$

To prove the more general case, we shall proceed with induction under the assumption that $\mathscr{L}\{t^nF(t)\} = (-1)^n \frac{d^n}{ds^n}f(s)$

$$\begin{aligned} \mathscr{L}\{t^nF(t)\} &= (-1)^n \frac{d^n}{ds^n}f(s) \\ \frac{d}{ds}\mathscr{L}\{t^nF(t)\} &= (-1)^n \frac{d^{n+1}}{ds^{n+1}}f(s) \\ \frac{d}{ds}\int_0^\infty t^nF(t)e^{-st}dt &= (-1)^n \frac{d^{n+1}}{ds^{n+1}}f(s) \\ \int_0^\infty t^{n+1}F(t)e^{-st}dt &= (-1)^n \frac{d^{n+1}}{ds^{n+1}}f(s) \end{aligned}$$

and the proof is complete. □

1.2 Exercises

Exercise 1.2 Determine from first principles the Laplace transform of the following functions:

(a) e^{kt}

(b) t^2

(c) $\cosh t$

Work I'm not doing this from first principles since it's kind of tedious and just integration by parts like twenty times so I will just be applying the theorems directly.

(a) Applying the First Shift Theorem 1.2 for $F(t) = 1$ we get

$$\mathscr{L}\left\{e^{kt}\right\} = \frac{1}{s-k}$$

(b) Applying Corollary 1.3 we get

$$\mathscr{L}\left\{t^2\right\} = \frac{2}{s^3}$$

(c)

$$\begin{aligned} \mathscr{L}\{\cosh t\} &= \frac{1}{2}\left(\mathscr{L}\left\{e^t\right\} + \mathscr{L}\left\{e^{-t}\right\}\right) = \frac{1}{2}\left(\frac{1}{s-1} + \frac{1}{s+1}\right) \\ &= \frac{2s}{2(s^2-1)} = \frac{s}{s^2-1} \end{aligned}$$

Exercise 1.3 Find the Laplace transforms of the following functions:

(a) t^2e^{-3t}

(b) $4t + 6e^{4t}$

(c) $e^{-4t}\sin(5t)$

Work

(a) Applying the First Shift Theorem 1.2 we get

$$\mathscr{L}\left\{t^2e^{-3t}\right\} = \frac{2}{(s+3)^3}$$

(b) Applying linearity we get

$$\mathscr{L}\left\{4t\right\} + 6\mathscr{L}\left\{e^{4t}\right\} = \frac{4}{s^2} + \frac{6}{s-4}$$

(c) Applying the First Shift Theorem 1.2 and subbing it into the Laplace Transform of a general exponential, we have

$$\mathscr{L}\left\{e^{-4t}\sin(5t)\right\} = \frac{5}{(s+4)^2+25}$$

Exercise 1.4 Find the Laplace Transform of the function $F(t)$ where $F(t)$ is given by

$$F(t) = \begin{cases} t & 0 \le t < 1 \\ 2-t & 1 \le t < 2 \\ 0 & \text{otherwise} \end{cases}$$

Work

$$\begin{aligned}
\mathscr{L}\left\{F(t)\right\} &= \int_0^1 te^{-st} + \int_1^2 (2-t)e^{-st}dt \\
&= \frac{1}{s^2} - \frac{s+1}{s^2}e^{-s} + \frac{e^{-2s}((s-1)e^s+1)}{s^2} \\
&= \frac{1-(s+1)e^{-s}+((s-1)e^s+1)e^{-2s}}{s^2} \\
&= \frac{1-(s+1)e^{-s}+e^{-2s}+(s-1)e^{-s}}{s^2} \\
&= \frac{e^{-2s}-2e^{-s}+1}{s^2} \\
&= \frac{(e^{-s}-1)^2}{s^2}
\end{aligned}$$

Exercise 1.5 Use the property of Theorem 1.5 to determine the following Laplace transforms

(a) te^{2t}

(b) $t\cos(t)$

(c) $t^2\cos(t)$

Work

(a)

$$\mathscr{L}\left\{te^{2t}\right\} = -\frac{d}{ds}\frac{1}{s-2} = \frac{1}{(s-2)^2}$$

(b)

$$\mathscr{L}\left\{t\cos t\right\} = -\frac{d}{ds}\frac{s}{s^2+1} = -\frac{s^2+1-2s^2}{(s^2+1)^2} = \frac{s^2-1}{(s^2+1)^2}$$

(c)

$$\mathscr{L}\left\{t^2\cos t\right\} = -\frac{d}{ds}\frac{s^2-1}{(s^2+1)^2} = \frac{2s(s^2+1)^2-(s^2-1)2(s^2+1)2s}{(s^2+1)^4} = \frac{2s^3-6s}{(s^2+1)^3}$$

Exercise 1.6 Find the Laplace transforms of the following functions:

(a) $\sin(\omega t+\phi)$

(b) $e^{5t}\cos t$

Work

(a)

$$\mathscr{L}\left\{\sin(\omega t+\phi)\right\} = \int_0^\infty e^{-st}\sin(\omega t+\phi)dt = \frac{s\sin\phi+\omega\cos\phi}{\omega^2+s^2}$$

(b)

$$\mathscr{L}\left\{e^{5t}\cosh t\right\} = \frac{s}{(s-5)^2-6^2}$$

Exercise 1.7 If $G(at+b)=F(t)$, determine the Laplace transform of G in terms of $\mathscr{L}\{F\}=\bar{f}(s)$ and a finite integral.

Work Let us begin by considering the substitution $t=au+b$ so that $dt=adu$ for the integral

$$\mathscr{L}\{G(t)\} = \int_0^\infty e^{-st}G(t)dt$$

We therefore have that $G(t)=G(au+b)=F(u)$ and the above equation equals

$$\begin{aligned}\int_{-b/a}^\infty ae^{-s(au+b)}F(u)du &= \int_{-b/a}^\infty ae^{-aus}e^{-sb}F(u)du\\ &= ae^{-sb}\int_{-b/a}^\infty e^{-asu}F(u)du\\ &= ae^{-sb}\int_0^\infty e^{-asu}F(u)du + ae^{-sb}\int_{-b/a}^0 e^{-asu}F(u)du\\ &= ae^{-sb}\bar{f}(as) + ae^{-sb}\int_{-b/a}^0 e^{-asu}F(u)du\end{aligned}$$

Exercise 1.8 Prove the following change of scale result:

$$\mathscr{L}\{F(at)\} = \frac{1}{a}f\left(\frac{s}{a}\right)$$

Hence evaluate the Laplace transforms of the two functions:

(a) $t\cos(6t)$

(b) $t^2\cos(7t)$

Work

Proof. The proof proceeds by making a substitution such that $u=at\rightarrow du=adt$

$$\mathscr{L}\{F(at)\} = \int_0^\infty e^{-st}F(at)dt = \int_0^\infty e^{-s/a}F(u)\frac{1}{a}du = \frac{1}{a}f\left(\frac{s}{a}\right)$$

□

(a)

$$\begin{aligned}\mathscr{L}\{t\cos(6t)\} &= -\frac{d}{ds}\frac{1}{6}\frac{s/6}{(s/6)^2+1} = -\frac{d}{ds}\frac{s}{s^2+36} \\ &= -\frac{s^2+36-s(2s)}{(s^2+36)^2} = -\frac{-s^2+36}{(s^2+36)^2} \\ &= \frac{s^2-36}{(s^2+36)^2}\end{aligned}$$

(b)

$$\begin{aligned}\mathscr{L}\left\{t^2\cos(7t)\right\} &= -\frac{d}{ds}\frac{s^2-49}{(s^2+49)^2} \\ &= -\frac{2s(s^2+49)^2-4s(s^2+49)(s^2-49)}{(s^2+49)^4} \\ &= \frac{2s(s^2-147)}{(s^2+49)^3}\end{aligned}$$

Chapter 2

Further Properties of the Laplace Transform

2.1 Derivative Property

Theorem 2.1 *Suppose a differentiable function $F(t)$ has a Laplace transform $f(s)$. Then the Laplace transform of its derivative $F'(t)$ is*

$$\mathscr{L}\{F'(t)\} = \int_0^\infty F'(t)e^{-st}dt = -F(0) + sf(s)$$

Proof. We can easily prove this by integrating by parts once

$$\begin{aligned}\mathscr{L}\{F'(t)\} &= \int_0^\infty F'(t)e^{-st}dt \\ &= F(t)e^{-st}\Big|_0^\infty + s\int_0^\infty F(t)e^{-st}dt \\ &= -F(0) + sf(s)\end{aligned}$$

□

We can state and prove the more general case of this property by using induction.

Theorem 2.2 *Suppose $F(t) \in C^n$, then the Laplace transform of its nth derivative is*

$$\mathscr{L}\left\{F^{(n)}(t)\right\} = s^n f(s) - s^{n-1}F(0) - s^{n-2}F'(0) - \ldots - F^{(n-1)}(0) \tag{2.1}$$

Proof. Let us start with the definition given in (2.1) and perform some algebra

$$\begin{aligned}\mathscr{L}\left\{F^{(n)}(t)\right\} &= s^n f(s) - s^{n-1}F(0) - s^{n-2}F'(0) - \ldots - F^{(n-1)}(0) \\ &= s\left(s^{n-1}f(s) - s^{n-2}F(0) - s^{n-3}F'(0) - \ldots - F^{(n-2)}(0)\right) - F^{(n-1)}(0) \\ &= s\mathscr{L}\left\{F^{(n-1)}(t)\right\} - F^{(n-1)}(0)\end{aligned}$$

which we can verify easily by differentiating and, assuming that $F(t) \in C^{n+1}$, we can move the differentiation operator into the integral and perform an integration by parts to achieve our desired result

$$\begin{aligned}\frac{d}{dt}\mathscr{L}\left\{F^{(n)}(t)\right\} &= \mathscr{L}\left\{F^{n+1)}(t)\right\} \\ &= \int_0^\infty F^{(n+1)}(t)e^{-st}dt \\ &= F^{(n)}(t)e^{-st}\Big|_0^\infty + s\int_0^\infty F^{(n)}(t)e^{-st}dt \\ &= s\mathscr{L}\left\{F^{(n)}(t)\right\} - F^{(n)}(0)\end{aligned}$$

Now we just need to verify the base case for $n = 2$ which is easily proven by integrating by parts twice. □

This can be used to prove a result we proved previously. We know from before that the Laplace transform of $\sin(\omega t) = \frac{\omega^2}{s^2+\omega^2}$ and we know that $\frac{d}{dt}(\sin(\omega t)) = \omega\cos(\omega t)$ and so thus

$$\mathscr{L}\{\omega\cos(\omega t)\} = s\frac{\omega}{s^2+\omega^2} - 0 \to \mathscr{L}\{\cos(\omega t)\} = \frac{s}{s^2+\omega^2}$$

We can use this theorem to find the value of the function

$$g(t) = \int_0^t F(u)du$$

assuming that $F(t)$ is integrable and $g(t)$ is is of exponential order. We can see easily that $g(0) = 0$ and $g'(t) = F(t)$ by the Fundamental Theorem of Calculus.

$$\begin{aligned}\mathscr{L}\{g'(t)\} = s\bar{g}(s) - g(0) = f(s) = \mathscr{L}\{F(t)\}\\ \Rightarrow \bar{g}(s) = \frac{f(s)}{s}\\ \Rightarrow \mathscr{L}\left\{\int_0^t F(u)du\right\} = \frac{f(s)}{s}\end{aligned}$$

Theorem 2.3 *If $\mathscr{L}\{F(t)\} = f(s)$, then $\mathscr{L}\left\{\frac{F(t)}{t}\right\} = \int_s^\infty f(u)du$ assuming that*

$$\mathscr{L}\left\{\frac{F(t)}{t}\right\} \to \text{ as } s \to \infty$$

Proof. Let $G(t)$ be the function $\frac{F(t)}{t}$ so that $F(t) = tG(t)$. Recalling Theorem 1.5, we have that

$$\mathscr{L}\{F(t)\} = f(s) = -\frac{d}{ds}\mathscr{L}\left\{\frac{F(t)}{t}\right\}$$

Integrating both sides from s to ∞, we get

$$\int_s^\infty f(u)du = -\mathscr{L}\left\{\frac{F(t)}{t}\right\}\Big|_0^\infty = 0 + \mathscr{L}\left\{\frac{F(t)}{t}\right\}$$

under the assumption that

$$\mathscr{L}\left\{\frac{F(t)}{t}\right\} \to 0 \text{ as } s \to \infty$$

□

Let us note that these two results above are the analogs of the derivative property in (2.1) and Theorem 1.5. That is to say, an integration in the s domain is a division in the t domain and vice versa.

Now let $F(t) = \sin t$. Applying the result from Theorem 2.3, we get that

$$\begin{aligned}\mathscr{L}\left\{\frac{\sin t}{t}\right\} &= \int_s^\infty \frac{du}{1+u^2}\\ &= \tan^{-1}(u)\Big|_s^\infty\\ &= \frac{\pi}{2} - \tan^{-1}(s)\\ &= \tan^{-1}\left(\frac{1}{s}\right)\end{aligned}$$

This is done by applying the identity

$$\tan^{-1}\left(\frac{1}{s}\right) + \tan^{-1}(s) = \frac{\pi}{2}$$

which we informally prove down below

$$\begin{aligned} f(x) &= \tan^{-1} x + \tan^{-1} \frac{1}{x} \\ \Rightarrow \tan(f(x)) &= \tan\left(\tan^{-1} x + \tan^{-1} \frac{1}{x}\right) \\ &= \frac{\tan(\tan^{-1} x) + \tan(\tan^{-1} \frac{1}{x})}{1 - (\tan(\tan^{-1} x))(\tan(\tan^{-1} \frac{1}{x}))} \\ &= \frac{x + \frac{1}{x}}{1 - 1} \end{aligned}$$

Since $\tan(f(x))$ is undefined for all x, then $f(x)$ must be constant. We know that tangent is always undefined when its argument is $\pm\frac{\pi}{2}$ and if $x > 0$, then we know the output must be positive so the constant is $\frac{\pi}{2}$.

Now let us study the Sine Integral function which is defined as

$$\mathrm{Si}(t) = \int_0^t \frac{\sin u}{u} du$$

We use the previous result $\mathscr{L}\left\{\frac{\sin t}{t}\right\} = \tan^{-1}\frac{1}{s}$ to show that the Laplace transform of the Sine Integral function is

$$\mathscr{L}\left\{\mathrm{Si}(t)\right\} = \mathscr{L}\left\{\int_0^t \frac{\sin u}{u} du\right\} = \frac{1}{s}\mathscr{L}\left\{\frac{\sin t}{t}\right\} = \frac{1}{s}\tan^{-1}\frac{1}{s}$$

2.2 The Heaviside Step Function

The Heaviside Unit Step function is defined as

$$H(t) = \begin{cases} 0 & x < 0 \\ 1 & x \geq 0 \end{cases}$$

The Laplace transform of the shifted unit step function is as follows

$$\mathscr{L}\left\{H(t-t_0)\right\} = \int_{t_0}^{\infty} e^{-st} dt = -\frac{e^{-st}}{s}\bigg|_0^{\infty} = \frac{e^{-st_0}}{s}$$

Now let us prove the Second Shift Theorem.

Theorem 2.4 (Second Shift Theorem) *If $F(t)$ is a function of exponential order in t and the Laplace transform of $\mathscr{L}\{F(t)\} = f(s)$ then*

$$\mathscr{L}\left\{H(t-t_0)F(t-t_0)\right\} = e^{-st_0} f(s)$$

Proof.

$$\begin{aligned} \mathscr{L}\left\{H(t-t_0)F(t-t_0)\right\} &= \int_0^{\infty} H(t-t_0)F(t-t_0)e^{-st} dt \\ &= \int_{t_0}^{\infty} F(t-t_0)e^{-st} dt \\ &= \int_0^{\infty} F(u)e^{-s(u+t_0)} du \\ &= e^{-st_0}\int_0^{\infty} F(u)e^{-su} du \\ &= e^{-st_0} f(s) \end{aligned}$$

where we have made the substitution $u = t - t_0$ such that $du = dt$. Note, it is important that when you are applying the Second Shift theorem that the function $F(t)$ must have the SAME argument as $H(t)$. By this, I mean that if $H(t-t_0)$, the argument of $F(\tau)$ MUST be $\tau = t - t_0$. □

2.3 Inverse Laplace Transforms

Definition 2.5 If $F(t)$ has the Laplace transform $f(s)$, then the inverse Laplace transform is uniquely defined (with the exception of functions whose Laplace transforms are 0) by

$$\mathscr{L}^{-1}\{f(s)\} = F(t)$$

Theorem 2.6 *The inverse Laplace transform is linear.*

Proof. We have already established the linearity of Laplace transform

$$\mathscr{L}\{aF_1(t) + bF_2(t)\} = af_1(s) + bf_2(s)$$

Applying the inverse gives

$$aF_1(t) + bF_2(t) = \mathscr{L}^{-1}\{af_1(s) + bf_2(s)\} = a\mathscr{L}^{-1}\{f_1(s)\} + b\mathscr{L}^{-1}\{f_2(s)\}$$

and therefore the inverse Laplace transform is linear. □

In this present chapter, there is no systematic way to find the Inverse Laplace transform. The way to do this requires complex analysis and will not be touched on until Chapter 8. Inverting is therefore reliant on recognizing the forms of known Laplace transforms.

2.4 Limiting Theorems

In the study of differential equations, the behaviors of systems for large values of t or s is an important part of the study of asymptotics. We shall now prove some limiting theorems.

Theorem 2.7 (Initial Value) *If the indicated limits exist, then*

$$\lim_{t\to 0} F(t) = \lim_{s\to\infty} sf(s)$$

where the left hand side is $F(0)$ *or* $\lim_{t\to 0^+} F(t)$ *if* $\lim_{t\to 0} F(t)$ *isn't unique.*

Proof. We start by using Theorem 2.1. Assuming $F'(t)$ is of exponential order and is piecewise continuous, we then see

$$\begin{aligned}\left|\int_0^\infty e^{-st}F'(t)dt\right| &\le \int_0^\infty |e^{-st}F'(t)|dt \\ &\le \int_0^\infty e^{-st}e^{Mt}dt \\ &= -\frac{1}{M-s}\end{aligned}$$

where we can bring the absolute value bars inside the integral through the triangle inequality. We therefore have that

$$\mathscr{L}\{F'(t)\} = sf(s) - F(0) \le -\frac{1}{M-s}$$

and therefore as $s\to\infty$, $\mathscr{L}\{F'(t)\}\to 0$ and we can rearrange to get

$$\lim_{s\to\infty} sf(s) = F(0) = \lim_{t\to 0} F(t)$$

□

Theorem 2.8 (Final Value) *If the limits indicated exist, then*

$$\lim_{t\to\infty} F(t) = \lim_{s\to 0} sf(s)$$

Proof. We once again start with result from Theorem 2.1 but we write out its integral form

$$\mathscr{L}\{F'(t)\} = \int_0^\infty e^{-st}F'(t)dt = sf(s) - F(0) \tag{2.2}$$

Now we take the limit as $s \to 0$

$$\begin{aligned}\lim_{s\to 0}\int_0^\infty e^{-st}F'(t)dt &= \lim_{s\to 0}\lim_{T\to\infty}\int_0^T e^{-st}F'(t)dt \\ &= \lim_{s\to 0}\lim_{T\to\infty}[e^{-sT}F(T) = F(0)] \\ &= \lim_{T\to\infty}F(T) - F(0)\end{aligned}$$

We have swapped the limits under the assumption that taking the s limit first yields an equivalent value to taking the T limit first and taking both limits at the same time which is a result from real analysis. Equating this with (2.2), we get

$$\lim_{t\to\infty}F(t) - F(0) = \lim_{s\to 0}sf(s) - F(0)$$

and the theorem is proved. □

An interesting application of the linearity of the Laplace transform is if $F(t)$ is an analytic function (as in we can represent it using a power series), we can effectively approximate the Laplace transform of $F(t)$ by multiplying each of the coefficients with the Laplace transform of $\mathscr{L}\{t^n\}$ which we already know the form of.

2.5 The Dirac-δ Function

I will now copy my notes on the delta function from my readings of Peter Olver's *Introduction to Partial Differential Equations* because firstly, I think his book is so good and secondly, he introduces two different interpretations of the delta function that are nice to think about.

There are two interpretations of the delta function: one as the limit of a series of functions and one as a linear functional. This is because the delta function isn't a function in the traditional but "akshually a distribution" which I've heard about 5 million times from every Youtube video that mentions a delta function.

In the limit interpretation, we choose a sequence of smooth functions $g_n(x)$ and require the limit $\lim_{n\to\infty} g_n(x) = 0\ \forall x \neq \xi$ where ξ is the location we want the impulse to be located. The other requirement is that $\int_a^b g_n(x)dx = 1\ \forall n \in \mathbb{N}$. As such, $\delta_\xi(x) = \lim_{n\to\infty} g_n(x)$. One such example is $g_n(x) = \frac{n}{\pi(1+n^2x^2)}$.

The other interpretation is the delta function as a linear functional. The most important property of the delta function is the sifting property.

$$\int_a^b \delta_\xi(x)u(x)dx = \int_a^b \delta_\xi(x)u(\xi)dx = u(\xi)\int_a^b \delta_\xi(x)dx = u(\xi) \text{ for } a < \xi < b$$

We can define the delta function in this framework as a linear function $L_\xi : C^0[a,b] \mapsto \mathbb{R}$ that maps a continue function $u(x)$ to its value at $x = \xi$. In a finite dimensional vector space, the operator $L : \mathbb{R}^n \mapsto \mathbb{R}$ is done by taking the dot product between the two. In an infinite dimensional function space, this is done by the inner product. In other words,

$$L_g[u] = \langle g, u\rangle = \int_a^b g(x)u(x)dx$$

The problem with this with regards to the delta function is that there does not exist a conventional function that satisfies the sifting property. Generalized functions are linear functionals acting on a function space but are intuitively a function through the L^2 inner product.

Following the limit interpretation, let us take the series of Top Hat functions defined as

$$T_p(t) = \begin{cases} 0 & t \leq -\frac{1}{T} \\ \frac{1}{2}T & -\frac{1}{T} < t < \frac{1}{T} \\ 0 & t \geq \frac{1}{T} \end{cases}$$

Assuming the integrals of the Top Hat function with a function $u(x)$ converges to $u(0)$, let us now delve into the Laplace transform of the delta function. Let us do this by taking the Laplace transform of the Top Hat function and then taking the limit as $T \to \infty$.

$$\begin{aligned}\mathscr{L}\{T_p(t)\} &= \int_0^\infty T_p(t)e^{-st}dt \\ &= \int_0^{\frac{1}{T}} \frac{1}{2}Te^{-st}dt \\ &= -\frac{T}{2s}e^{-st}\Big|_0^{\frac{1}{T}} \\ &= \frac{T}{2s} - \frac{T}{2s}e^{-\frac{s}{T}}\end{aligned}$$

For large values of T, we have that

$$e^{-\frac{s}{T}} \approx 1 - \frac{s}{T} + O\left(\frac{1}{T^2}\right)$$

We then see that the Laplace transform of the Top Hat function as $T \to \infty$ is

$$\frac{T}{2s} - \frac{T}{2s}e^{-\frac{s}{T}} \approx \frac{T}{2s} - \frac{T}{2s} + \frac{1}{2} + O\left(\frac{1}{T}\right) = \frac{1}{2} + O\left(\frac{1}{2}\right)$$

which goes to $\frac{1}{2}$ as $T \to \infty$. Usually in the study of Laplace transforms, we just define the Laplace transform of the delta function to be 1. We therefore use a different series of limiting functions to describe a delta function centered at $t = t_0$ using

$$K(t) = \begin{cases} 0 & t \leq t_0 - \frac{1}{2T} \\ \frac{1}{2}T & t_0 - \frac{1}{2T} < t < t_0 + \frac{1}{2T} \\ 0 & t \geq t_0 + \frac{1}{2T} \end{cases}$$

We can now once again use the sifting property to extract the value of a function at a specific time $t = t_0$. We can therefore find the Laplace transform of a function $\delta(t-a)f(t)$ to be

$$\mathscr{L}\{\delta(t-a)f(t)\} = e^{-as}f(a)$$

One interesting thing to note is the integral

$$\int_{-\infty}^{t} \delta(u - u_0)du$$

If $u_0 > t$, the integral evaluates to be 0. Otherwise, it is equal to 1. This is exactly the definition of the step function centered at u_0. This is by no means a rigorous proof of the fact that the delta function is the derivative of the step function, that is reserved for studies in the field of generalized functions. Rather, it is to gain intuition as to the relationships between these functions. We shall now state a property of nth derivatives of the delta function without proof since it is reserved for specialist texts.

$$\int_{-\infty}^{\infty} H(t)\delta^{(n)}(t)dt = (-1)^n h^{(n)}(0)$$

and

$$\int_{-\infty}^{\infty} e^{-st}\delta^{(n)}(t)dt = s^n$$

2.6 Periodic Functions

Definition 2.9 If $F(t)$ is a function that obeys the rule

$$F(t) = F(t + \tau)$$

for some $\tau > 0$ and $\tau \in \mathbb{R}$, then $F(t)$ is called a periodic function with period τ.

Theorem 2.10 *Let $F(t)$ have a period $T > 0$ such that $F(t) = F(t+T)$. Then*

$$\mathscr{L}\{F(t)\} = \frac{\int_0^T e^{-st}F(t)dt}{1-e^{-sT}}$$

Proof. We shall proceed by using the integral definition of the Laplace transform and recognize the periodicity of the integrand

$$\begin{aligned}\mathscr{L}\{F(t)\} &= \int_0^\infty e^{-st}F(t)dt \\ &= \int_0^T e^{-st}F(t)dt + \int_T^{2T} e^{-st}F(t)dt + \ldots \int_{(n-1)T}^{nT} e^{-st}F(t)dt + \ldots\end{aligned}$$

We now make the substitution $u = t-(n-1)T$ such that $du = dt$ such that the general integral becomes

$$\int_{(n-1)T}^{nT} e^{-st}F(t)dt = e^{-s(n-1)T}\int_0^T e^{-su}F(u)du \qquad n \in \mathbb{N}$$

Factoring the integral out of each term gives a geometric series with ratio e^{-sT}

$$\int_0^\infty e^{-st}F(t)dt = (1+e^{-sT}+e^{-2sT}+\ldots)\int_0^T e^{-st}F(t)dt = \frac{1}{1-e^{-sT}}\int_0^T e^{-st}F(t)dt$$

□

2.7 Exercises

Exercise 2.1 If $F(t) = \cos(at)$, use the derivative formula to re-establish the Laplace transform of $\sin(at)$.

Work

$$\begin{aligned}F(t) = \cos(at) \to F'(t) &= -a\sin(at) \\ \mathscr{L}\{F'(t)\} = \frac{s^2}{s^2+a^2} - 1 &= \frac{s^2-s^2-a^2}{s^2+a^2} = -\frac{a^2}{s^2+a^2} \\ \mathscr{L}\{\sin(at)\} &= -\frac{1}{a}\mathscr{L}\{F'(t)\} = \frac{a}{s^2+a^2}\end{aligned}$$

Exercise 2.2 Use Theorem 2.1 with

$$F(t) = \int_0^t \frac{\sin u}{u}du$$

to establish the result

$$\mathscr{L}\left\{\frac{\sin(at)}{t}\right\} = \tan^{-1}\left(\frac{a}{s}\right)$$

Work Let $F(t) = \displaystyle\int_0^t \frac{\sin u}{u}du$

$$\begin{aligned}F'(t) &= \frac{\sin at}{t} \\ \mathscr{L}\{F'(t)\} &= s\mathscr{L}\left\{\int_0^t \frac{\sin au}{u}du\right\} - F(0) \\ &= s\left(\tan^{-1}\left(\frac{s}{a}\right)\right)\frac{1}{s} - 0 \\ &= \tan^{-1}\left(\frac{s}{a}\right)\end{aligned}$$

where the Laplace transform of

$$\mathscr{L}\left\{\frac{\sin au}{u}\right\} = \int_s^\infty \frac{a}{a^2+v^2}dv$$
$$= \frac{\pi}{2} - a\frac{1}{a}\tan^{-1}\left(\frac{s}{a}\right)$$
$$= \tan^{-1}\left(\frac{a}{s}\right)$$

so now

$$\mathscr{L}\left\{\int_0^t \frac{\sin u}{u}du\right\} = \frac{1}{s}\tan^{-1}\left(\frac{a}{s}\right)$$

Exercise 2.3 Prove that

$$\mathscr{L}\left\{\int_0^t\int_0^v F(u)dudv\right\} = \frac{f(s)}{s^2}$$

Work

Proof. Let us define the function

$$G(v) = \int_0^v F(u)du$$

We can then take the Laplace transform of $G(v)$ to obtain

$$\mathscr{L}\{G(v)\} = \mathscr{L}\left\{\int_0^v F(u)du\right\} = \frac{f(s)}{s}$$

Now the Laplace transform of the integral of $G(v)$ is

$$\mathscr{L}\left\{\int_0^t G(v)dv\right\} = \frac{g(s)}{s} = \frac{f(s)}{s^2}$$

□

Exercise 2.4 Find

$$\mathscr{L}\left\{\int_0^t \frac{\cos(au)-\cos(bu)}{u}du\right\}$$

Work

$$\mathscr{L}\left\{\int_0^t \frac{\cos(au)-\cos(bu)}{u}du\right\} = \frac{1}{s}\mathscr{L}\left\{\frac{\cos(au)-\cos(bu)}{u}\right\}$$
$$= \frac{1}{s}\int_s^\infty\left(\frac{u}{u^2+a^2}-\frac{u}{u^2+b^2}\right)du$$

We now make the substitution $v = u^2 + a^2$ (or b^2) so now $dv = 2udu$

$$= \frac{1}{2s}\left(\int_{s^2+a^2}^\infty \frac{1}{v}dv - \int_{s^2+b^2}^\infty \frac{1}{v}dv\right)$$
$$= \frac{1}{2s}\int_{s^2+a^2}^{s^2+b^2}\frac{1}{v}dv$$
$$= \frac{1}{2s}\ln\left(\frac{s+b^2}{s+a^2}\right)$$

Exercise 2.5 Determine

$$\mathscr{L}\left\{\frac{2\sin t\sinh t}{t}\right\}$$

Work

$$\begin{aligned}
\mathscr{L}\left\{\frac{2\sin t\sinh t}{t}\right\} &= \int_s^\infty \mathscr{L}\left\{2\sin t\sinh t\right\} du \\
&= \int_s^\infty \mathscr{L}\left\{e^t\sin t - e^{-t}\sin t\right\} du \\
&= \int_s^\infty \left(\frac{1}{(s-1)^2+1} - \frac{1}{(s+1)^2+1}\right) du \\
&= \frac{\pi}{2} - \tan^{-1}(s-1) - \frac{\pi}{2} + \tan^{-1}(s+1) \\
&= \tan^{-1}\left(\frac{1}{s-1}\right) - \tan^{-1}\left(\frac{1}{s+1}\right) \\
&= \tan^{-1}\left(\frac{\frac{1}{s-1} - \frac{1}{s+1}}{1+\frac{1}{s^2-1}}\right) \\
&= \tan^{-1}\left(\frac{\frac{2}{s^2-1}}{\frac{s^2}{s^2-1}}\right) \\
&= \tan^{-1}\frac{2}{s^2}
\end{aligned}$$

Exercise 2.6 Prove that if $f(s)$ indicates the Laplace transform of a piecewise continuous function $F(t)$, then

$$\lim_{s\to\infty} f(s) = 0$$

Work

$$\lim_{s\to\infty} f(s) = \lim_{s\to\infty}\int_0^\infty e^{-st}F(t)dt = \int_0^\infty \lim_{s\to\infty} e^{-st}F(t)dt = \int_0^\infty 0dt = 0$$

Assuming the necessary convergence conditions, we can bring the limit into the integral (I think Lebesgue Dominated Convergence Theorem is used here but I'm unsure).

Exercise 2.7 Determine the following inverse Laplace transforms by using partial fractions:

(a) $\dfrac{2(2s+7)}{(s+4)(s+2)}$ for $s > -2$

(b) $\dfrac{s+9}{s^2-9}$

(c) $\dfrac{s^2+2k^2}{s(s^2+4k^2)}$

(d) $\dfrac{1}{s(s+3)^2}$

(e) $\dfrac{1}{(s-2)^2(s+3)^3}$

Work (a)

$$\begin{aligned}
\frac{2(2s+7)}{(s+4)(s+2)} &= \frac{1}{s+4} + \frac{3}{s+2} \\
\mathscr{L}^{-1}\left\{\frac{1}{s+4} + \frac{3}{s+2}\right\} &= e^{-4t} + 3e^{-2t}
\end{aligned}$$

(b)

$$\begin{aligned}
\frac{s+9}{s^2-9} &= \frac{2}{s-3} - \frac{1}{s+3} \\
\mathscr{L}^{-1}\left\{\frac{2}{s-3} - \frac{1}{s+3}\right\} &= 2e^{3t} - e^{-3t}
\end{aligned}$$

(c)

$$\frac{s^2+2k^2}{s(s^2+4k^2)} = \frac{1}{2s} + \frac{2}{(s^2+4k^2)}$$
$$\mathscr{L}^{-1}\left\{\frac{1}{2s} + \frac{2}{(s^2+4k^2)}\right\} = \frac{1}{2}\cos(2kt) + \frac{1}{2}$$

(d)

$$\frac{1}{s(s+3)^2} = \frac{1}{9s} - \frac{1}{9(s+3)} - \frac{1}{3(s+3)^2}$$
$$\mathscr{L}^{-1}\left\{\frac{1}{9s} - \frac{1}{9(s+3)} - \frac{1}{3(s+3)^2}\right\} = \frac{1}{9} - \frac{1}{9}e^{-3t} - \frac{1}{3}e^{-3t}t$$

(e)

$$\frac{1}{(s-2)^2(s+3)^3} = \frac{3}{625(s+3)} + \frac{2}{125(s+3)^2} + \frac{1}{25(s+3)^3} - \frac{3}{625(s-2)} + \frac{1}{125(s-2)^2}$$
$$\mathscr{L}^{-1}\left\{\frac{1}{(s-2)^2(s+3)^3}\right\} = \frac{3}{625}e^{-3t} + \frac{2}{125}te^{-3t} + \frac{1}{50}t^2e^{-3t} - \frac{3}{625}e^{2t} + \frac{1}{125}te^{2t}$$

Exercise 2.8 Verify the initial value theorem for the two functions

(a) $2 + \cos t$

(b) $(4+t)^2$

Work

(a) $\lim_{t\to 0} F(t) = 3 = \lim_{s\to\infty} \frac{2}{s} + s\left(\frac{s}{s^2+1}\right) = 3$

(b) $\lim_{t\to 0}(4+t)^2 = 16 = \lim_{s\to\infty} s\left(\frac{16}{s} + \frac{8}{s^2} + \frac{2}{s^3}\right) = 16$

Exercise 2.9 Verify the final value theorem for the two functions

(a) $3 + e^{-t}$

(b) t^3e^{-t}

Work

(a) $\lim_{t\to\infty} F(t) = 3 = \lim_{s\to 0} s\left(\frac{3}{s} + \frac{1}{s+1}\right) = 3$

(b) $\lim_{t\to\infty} t^3e^{-t} = 0 = \lim_{s\to 0} s\left(\frac{6}{(s+1)^4}\right) = 0$

Exercise 2.10 Given that

$$\mathscr{L}\left\{\sin(\sqrt{t})\right\} = \frac{k}{s^{3/2}}e^{-\frac{1}{4s}}$$

use $\sin x \approx x$ for small x to determine the value of the constant k.

Work Consulting a table of Laplace transforms, we see that

$$\mathscr{L}\left\{\frac{\sin 2\sqrt{at}}{\sqrt{\pi a}}\right\} = \frac{e^{-a/s}}{s\sqrt{s}}$$
$$\mathscr{L}\left\{\frac{\sin 2\sqrt{t/4}}{\sqrt{\pi/4}}\right\} = \frac{1}{\sqrt{\pi/4}}\mathscr{L}\left\{\sin\sqrt{t}\right\} = \frac{e^{-\frac{1}{4s}}}{s^{3/2}}$$
$$\Rightarrow k = \frac{\sqrt{\pi}}{2}$$

Exercise 2.11 By using a power series expansion, determine (in series form) the Laplace transforms of $\sin(t^2)$ and $\cos(t^2)$

Work

$$\sin x^2 = \sum_{n=0}^{\infty} \frac{(-1)^n x^{4n+2}}{(2n+1)!}$$
$$\cos x^2 = \sum_{n=0}^{\infty} \frac{(-1)^n x^{4n}}{(2n)!}$$
$$\mathscr{L}\left\{\sin x^2\right\} = \sum_{n=0}^{\infty} \frac{(-1)^n (4n+2)!}{(2n+1)! s^{4n+3}}$$
$$\mathscr{L}\left\{\cos x^2\right\} = \sum_{n=0}^{\infty} \frac{(-1)^n (4n)!}{(2n)! s^{4n+1}}$$

Exercise 2.12 $P(s)$ and $Q(s)$ are polynomials, the degree of $P(s)$ is less than that of $Q(s)$ which is n. Use partial fractions to prove the result

$$\mathscr{L}^{-1}\left\{\frac{P(s)}{Q(s)}\right\} = \sum_{k=1}^{n} \frac{P(a_k)}{Q'(a_k)} e^{a_k t}$$

where a_k are the n distinct zeros of $Q(s)$.

Work The function $\frac{P(s)}{Q(s)}$ can be expressed, when broken down completely by partial fractions, as

$$\frac{P(s)}{Q(s)} = \frac{A_1}{s-a_1} + \frac{A_2}{s-a_2} + \ldots + \frac{A_k}{s-a_k} + \ldots + \frac{A_n}{s-a_n}$$

Taking the limit as $s \to a_k$ after multiplying by $(s-a_k)$ $\forall k \in \{1, 2, \ldots, n\}$, we see that everything but the coefficient for the root a_k goes to zero.

$$\lim_{s \to a_k} \left((s-a_k) \frac{P(s)}{Q(s)} \right) = \frac{A_1(s-a_k)}{s-a_1} + \ldots + A_k + \ldots + \frac{A_n(s-a_k)}{s-a_n}$$

We then get by applying L'Hôpital's Rule

$$\begin{aligned} A_k &= \lim_{s \to a_k} \frac{P(s)}{Q(s)} (s-a_k) \\ &= P(a_k) \lim_{s \to a_k} \frac{s-a_k}{Q(s)} \\ &= P(a_k) \lim_{s \to a_k} \frac{1}{Q'(s)} \\ &= \frac{P(a_k)}{Q'(a_k)} \end{aligned}$$

We can therefore write $\frac{P(s)}{Q(s)}$ as

$$\frac{P(s)}{Q(s)} = \sum_{k=1}^{n} \frac{P(a_k)}{Q'(a_k)(s-a_k)}$$

and since A_k is just a constant, when we invert, we recognize the denominator is just the Laplace transform of 1 after applying the first shift theorem

$$\mathscr{L}^{-1}\left\{\frac{P(s)}{Q(s)}\right\} = \sum_{k=1}^{n} \frac{P(a_k)}{Q'(a_k)} e^{a_k t}$$

Exercise 2.13 Find the following Laplace transforms

(a) $H(t-a)$

(b) $f_1 = \begin{cases} t+1 & 0 \le t \le 2 \\ 3 & t > 2 \end{cases}$

(c) $f_2 = \begin{cases} t+1 & 0 \le t \le 2 \\ 6 & t > 2 \end{cases}$

Work

(a)

$$\mathscr{L}\{H(t-a)\} = \int_0^\infty H(t-a)e^{-st}dt = \int_a^\infty e^{-st}dt = \frac{e^{-as}}{s}$$

(b)

$$\begin{aligned} f_1(t) &= (t+1)\left[H(t) - H(t-2)\right] + 3H(t-2) \\ &= (t+1)H(t) - (t-2+3)H(t-2) + 3H(t-2) \\ &= (t+1)H(t) - (t-2)H(t-2) \\ \mathscr{L}\{f_1(t)\} &= \frac{1}{s^2} + \frac{1}{s} - e^{-2s}\frac{1}{s^2} \end{aligned}$$

(c) I guess I thought the rest were too tedious to do so I did not do them

Exercise 2.14 Find the Laplace transform of the triangular wave function

$$F(t) = \begin{cases} t & 0 \le t < c \\ 2c - t & c \le t < 2c \end{cases}$$
$$F(t+2c) = F(t)$$

Work

$$\begin{aligned} \mathscr{L}\{F(t)\} &= \frac{1}{1-e^{-2cs}}\left[\int_0^c e^{-st}dt + \int_c^{2c}(2c-t)e^{-st}dt\right] \\ &= \frac{1}{1-e^{-2cs}}\left[\frac{1}{s^2} - \frac{(cs+1)e^{-cs}}{s^2} + \frac{(cs-1)e^{-cs}}{s^2} + \frac{e^{-2cs}}{s^2}\right] \\ &= \frac{1}{1-e^{-2cs}}\left[1 - 2e^{-cs} + e^{-2cs}\right] \\ &= \frac{1}{1-e^{-2cs}}\left[1 - e^{-cs}\right]^2 \\ &= \frac{1}{s^2}\frac{1-e^{-sc}}{1+e^{-sc}} \\ &= \frac{1}{s^2}\tanh\left(\tfrac{1}{2}sc\right) \end{aligned}$$

Exercise 2.15 Find the Laplace transform of the generally placed top hat function

$$F(t) = \begin{cases} \frac{1}{h} & a \le t < a+h \\ 0 & \text{otherwise} \end{cases}$$

Hence deduce the Laplace transform of the Dirac-δ function $\delta(t-a)$ where $a > 0$ is a real constant.

Work

$$\begin{aligned}
\mathscr{L}\{F(t)\} &= \int_a^{a+h} \frac{1}{h} e^{-st} dt \\
&= -\frac{1}{sh} e^{-st} \Big|_a^{a+h} \\
&= -\frac{1}{hs} e^{-(a+h)s} + \frac{1}{hs} e^{-as} \\
&= \frac{1}{hs} e^{-as}(1 - e^{-hs}) \\
\Rightarrow \lim_{h\to 0} \frac{(1-e^{-hs})e^{-as}}{hs} &= \frac{e^{-as}}{s} \lim_{h\to 0} \frac{1-e^{-hs}}{h} \\
&= \frac{e^{-as}}{s} \frac{s}{1} \\
&= e^{-as}
\end{aligned}$$

Chapter 3

Convolution and Ordinary Differential Equations

3.1 Convolution

Perhaps one of the most important mathematical operations is convolution. Convolution has many applications, particularly in electrical engineering where it's fundamental to digital signal processing. The convolution operator is defined as follows.

Definition 3.1 (Convolution) The convolution of two given functions $f(t)$ and $g(t)$ is written as $f * g$ and is defined by

$$f * g = \int_0^t f(\tau)g(t-\tau)d\tau$$

The only requirement is that the integral on the right exists.

Theorem 3.2 (Symmetry) *The convolution operator is symmetric. That is to say,*

$$f * g = g * f$$

Proof. The proof is simple under the substitution $u = t - \tau \to du = -d\tau$

$$f * g = \int_0^t f(\tau)g(t-\tau)d\tau = -\int_t^0 f(t-u)g(u)du = \int_0^t f(t-u)g(u)du = g * f$$

□

Theorem 3.3 (Convolution) *If $F(t)$ and $G(t)$ are of exponential order and $\mathscr{L}\{F(t)\} = \bar{f}(s)$ and $\mathscr{L}\{G(t)\} = \bar{g}(s)$ are their Laplace transforms, then $\mathscr{L}^{-1}\left\{\bar{f}\bar{g}\right\} = f * g$.*

Proof. We show via direct integration that

$$\bar{f}\bar{g} = \mathscr{L}\{F(t) * G(t)\}$$

Taking the Laplace transform of the convolution integral yields

$$\mathscr{L}\{F * G\} = \int_0^\infty e^{-st}\int_0^t F(\tau)G(t-\tau)d\tau dt$$

Let us note in the t-τ plane that the integration domain is from the t-axis to the 45° line $t = \tau$. This allows us to change the bounds of integration such that the t integral ranges from $\tau \to \infty$ and the τ integral ranges from $0 \to \infty$.

$$\mathscr{L}\{F * G\} = \int_0^\infty F(\tau)\int_\tau^\infty G(t-\tau)e^{-st}dtd\tau$$

For the inner integral, let us make the substitution $u = t - \tau \to du = dt$

$$\begin{aligned}\int_\tau^\infty G(t-\tau)e^{-st}dt &= \int_0^\infty G(u)e^{-s(u+\tau)}du \\ &= e^{-s\tau}\int_0^\infty G(u)e^{-su}du \\ &= e^{-s\tau}\bar{g}(s)\end{aligned}$$

The double integral then simplifies down to

$$\begin{aligned}\int_0^\infty F(\tau)\int_\tau^\infty G(t-\tau)e^{-st}dtd\tau &= \int_0^\infty F(\tau)\bar{g}(s)e^{-s\tau}d\tau \\ &= \bar{f}(s)\bar{g}(s) \\ \Rightarrow \mathscr{L}^{-1}\left\{\bar{f}(s)\bar{g}(s)\right\} &= F(t) * G(t)\end{aligned}$$

□

Let us find the Laplace transform of $\frac{1}{\sqrt{t}}$ as it prove useful later. We start with the substitution $st = u^2 \to 2udu = sdt$

$$\mathscr{L}\left\{\frac{1}{\sqrt{t}}\right\} = \int_0^\infty \frac{e^{-u^2}}{s}\frac{\sqrt{s}}{u}2udu = \frac{2}{\sqrt{s}}\int_0^\infty e^{-u^2}du = \sqrt{\frac{\pi}{s}}$$

Where we recognize that

$$\int_0^\infty e^{-x^2}dx = \frac{\sqrt{\pi}}{2}$$

as the Gaussian integral. We can also see that

$$\mathscr{L}^{-1}\left\{\frac{1}{\sqrt{s}}\right\} = \frac{1}{\sqrt{\pi t}}$$

Example 3.4 We shall now find the inverse Laplace transform of

$$\mathscr{L}^{-1}\left\{\frac{1}{\sqrt{s}(s-1)}\right\}$$

Let us recognize that these are two known transforms and apply the convolution theorem (3.3)

$$\begin{aligned}\mathscr{L}^{-1}\left\{\frac{1}{\sqrt{s}(s-1)}\right\} &= \frac{1}{\sqrt{\pi t}} * e^t \\ &= \int_0^t \frac{1}{\sqrt{\pi t}}e^{t-\tau}d\tau \\ &= \frac{e^t}{\sqrt{\pi}}\int_0^t \frac{e^{-\tau}}{\sqrt{\tau}}d\tau\end{aligned}$$

Now we make the substitution $u^2 = \tau \to 2udu = d\tau$

$$\begin{aligned}\frac{e^t}{\sqrt{\pi}}\int_0^{\sqrt{t}}\frac{e^{-u^2}}{u}2udu &= \frac{2e^t}{\sqrt{\pi}}\int_0^{\sqrt{t}}e^{-u^2}du \\ &= e^t\,\mathrm{erf}\left(\sqrt{t}\right)\end{aligned}$$

where the error function $\mathrm{erf}\,(x)$ is defined as

$$\mathrm{erf}(x) = \frac{2}{\sqrt{\pi}}\int_0^x e^{-t^2}dt$$

and the complementary error function is defined as $1 - \mathrm{erf}(x)$.

Example 3.5 Another important result comes from finding the Laplace transform of

$$\mathscr{L}\left\{t^{-\frac{3}{2}}e^{-\frac{k^2}{4t}}\right\}$$

Let us start by making the substitution $u = \frac{k}{2\sqrt{t}} \to du = -\frac{k}{4}t^{-\frac{3}{2}}dt$ and then completing the square to get

$$\begin{aligned}\mathscr{L}\left\{t^{-\frac{3}{2}}e^{-\frac{k^2}{4t}}\right\} &= \int_0^\infty e^{-k^2/4t-st}t^{-3/2}dt \\ &= \frac{4}{k}\int_0^\infty e^{-u^2-sk/4u^2}du \\ &= \frac{4}{k}\int_0^\infty e^{-(u-\frac{k\sqrt{s}}{2u})^2-k\sqrt{s}}du\end{aligned}$$

We then express $u - \frac{k\sqrt{s}}{2u}$ as $u - \frac{u}{a}$. We then make the substitution $v = \frac{a}{u} \to dv = -\frac{a}{u^2}du$ and also note the identity $(v - \frac{a}{v})^2 = (\frac{a}{u} - u)^2$ and that the minus sign from the differential cancels the limit change so we get

$$\frac{4}{k}\int_0^\infty e^{-(u-\frac{a}{u})^2-k\sqrt{s}}du = \frac{4}{k}e^{-k\sqrt{s}}\int_0^\infty \frac{a}{u^2}e^{-(u-\frac{a}{u})^2}du$$

after replacing v with u. We then add the left hand integral to both sides to reach

$$\frac{8}{k}e^{-k\sqrt{s}}\int_0^\infty \frac{a}{u^2}e^{-(u-\frac{a}{u})^2}du = \frac{4}{k}e^{-k\sqrt{s}}\int_0^\infty \left(1+\frac{a}{u^2}\right)e^{-(u-\frac{a}{u})^2}du$$

We now make the substitution $\lambda = u - \frac{a}{u} \to d\lambda = \left(1 + \frac{a}{u^2}\right)du$

$$\begin{aligned}\frac{4}{k}e^{-k\sqrt{s}}\int_0^\infty \left(1+\frac{a}{u^2}\right)e^{-(u-\frac{a}{u})^2}du &= \frac{4}{k}e^{-k\sqrt{s}}\int_{-\infty}^\infty e^{-\lambda^2}d\lambda \\ &= \frac{4}{k}e^{-k\sqrt{s}}\sqrt{\pi} \\ \Rightarrow \mathscr{L}\left\{t^{-\frac{3}{2}}e^{-\frac{k^2}{4t}}\right\} &= \frac{4}{k}e^{-k\sqrt{s}}\int_0^\infty e^{-(u-\frac{u}{a})^2}du \\ &= \frac{2\sqrt{\pi}}{k}e^{-k\sqrt{s}} \\ \Rightarrow \mathscr{L}^{-1}\left\{e^{-k\sqrt{s}}\right\} &= \frac{k}{2\sqrt{\pi t^3}}e^{-\frac{k^2}{4t}}\end{aligned}$$

3.2 Ordinary Differential Equations

We shall consider the general second order differential equation

$$a\frac{d^2x}{dt^2} + b\frac{dx}{dt} + cx = f(x) \qquad (t \geq 0)$$

For the example of a mass on a spring, the second derivative term represents the acceleration, the first derivative term represents the damping, x represents the displacement term, and $f(x)$ represents the forcing term. Since the coefficients are constant, the system is deemed "time invariant." This means a response to a specific stimulus will be the same regardless at what time t it happens. An example of a time invariant system can be found by finding the general solution to the heat equation which can be read upon further here in Section 3.

Taking the Laplace transform of both sides gives

$$a(s^2\bar{x}(s) - sx(0) - x'(0)) + b(s\bar{x}(s) - x(0)) + c\bar{x}(s) = \bar{f}(s)$$

Solving for $\bar{x}(s)$ yields

$$\bar{x}(s) = \frac{\bar{f}(s) + (as+b)x(0) + ax'(0)}{as^s + bs + c}$$

If we assume homogeneous initial conditions ($x(0) = x'(0) = 0$), we achieve

$$\bar{x}(s) = \frac{1}{as^s + bs + c}\bar{f}(s)$$

which is of the form "response = transfer function $\times$ input." The transfer function maps every an input to an output and so intuitively is similar to that of a Green's function. Let us look at an example of a differential equation being solved using Laplace transforms.

Example 3.6 The equation governing the bending of beams is of the form

$$k\frac{d^4y}{dx^4} = -W(x)$$

where k is flexural rigidity constant (the product of the Young's modulus and the length), and $W(x)$ is the transverse force per unit length along the beam.

Let us first find the general solution for a cantilever (where one side is supported and the other end is free). Since the beam is finite, let us write the Laplace transform of the $y(x)$, the function that describes the deflection of the beam, as

$$\bar{y}(s) = \int_0^\infty y(x)[1 - H(x-l)]e^{-xs}dx$$

Taking Laplace transforms of the fourth order differential equation yields

$$k(s^4\bar{y}(s) - s^3y(0) - s^2y'(0) - sy''(0) - y'''(0)) = -\overline{W}(s)$$
$$\bar{y}(s) = \frac{-\overline{W}(s)}{ks^4} + \frac{y(0)}{s} + \frac{y'(0)}{s^2} + \frac{y''(0)}{s^3} + \frac{y'''(0)}{s^4}$$

Applying the convolution theorem 3.3 to find

$$\mathscr{L}^{-1}\left\{\frac{\overline{W}(s)}{s^4}\right\} = \frac{1}{6}\int_0^x (x-\xi)^3W(\xi)d\xi$$

We then have that the general solution to the cantilever problem to be

$$y(x)[1 - H(x-l)] = -\frac{1}{6k}\int_0^x (x-\xi)^3W(\xi)d\xi + y(0) + xy'(0) + \frac{1}{2}x^2y''(0) + \frac{1}{6}x^3y'''(0)$$

Let us now look at the case in which both ends are supported. This implies the boundary conditions that there is no displacement at either end ($y(x) = y(l) = 0$) and that there is no net force there either ($y''(0) = y''(l) = 0$). We can apply the boundary conditions at $x = 0$ to arrive at

$$y(x)[1 - H(x-l)] = -\frac{1}{6k}\int_0^x (x-\xi)^3W(\xi)d\xi + xy'(0) + \frac{1}{6}x^3y'''(0)$$

It is non-trivial to substitute in the $x = l$ boundary conditions. One method is to differentiate with respect to x twice but we do not know how the product of $y(x)$ and the step function behaves under differentiation. As such, let us substitute $u(x) = y''(x)$ so now

$$k\frac{d^2u}{dx^2} = -W(x)$$

Applying the Laplace transform and doing the same as above, we arrive at

$$u(x)[1 - H(x-l)] = -\frac{1}{k}\int_0^x (x-\xi)W(\xi)d\xi + u(0) + xu'(0)$$
$$\Rightarrow y''(x)[1 - H(x-l)] = -\frac{1}{k}\int_0^x (x-\xi)W(\xi)d\xi + y''(0) + xy'''(0)$$

Substituting in the boundary conditions at $x = l$ and rearranging, we get

$$y'''(0) = -\frac{1}{kl}\int_0^l (l-\xi)W(\xi)d\xi$$

Finding the last initial condition is

$$0 = -\frac{1}{6k}\int_0^l (l-\xi)^3 W(\xi)d\xi + ly'(0) + \frac{l^3}{6}\frac{1}{kl}\int_0^l (l-\xi)W(\xi)d\xi$$
$$\Rightarrow y'(0) = \frac{1}{6kl}\int_0^l (l-\xi)^3 W(\xi)d\xi - \frac{l}{6k}\int_0^l (l-\xi)W(\xi)d\xi$$

Therefore the general solution is

$$\begin{aligned} y(x)[1-H(x-l)] = &-\frac{1}{6k}\int_0^x (x-\xi)^3 W(\xi)d\xi + \frac{x}{6kl}\int_0^l (l-\xi)^3 W(\xi)d\xi \\ &-\frac{xl}{6k}\int_0^l (l-\xi)W(\xi)d\xi + \frac{x^3}{6kl}\int_0^l (l-\xi)W(\xi)d\xi \end{aligned}$$

From this, we can now find the solution to any weight distribution $W(x)$.

3.3 Integral Equations

Integral equations are equations where the unknown function appears under an integral sign. One example is

$$\int_a^b K(x,y)\phi(y)dy = f(x)$$

where $K(x,y)$ is known as the kernel and is known along with $f(x)$. One can view this as an integral operator acting on an unknown $\phi(y)$ leading to $f(x)$. Another integral equation is

$$A(x)\phi(x) - \lambda\int_a^b K(x,y)\phi(y)dy = f(x)$$

which is known as a Fredholm integral of the third kind. If $A(x) = 1$, then that is known as a Fredholm equation of the second kind. In Fredholm integrals, a and b are constant but when a or b are dependent on the independent variable x, then that is known as a Volterra integral equation. An integral equation that is easy to solve with Laplace transforms is the case when $b = x$. Let us look at an example.

Example 3.7 Solve the integral equation

$$\phi(x) - \lambda\int_0^x e^{x-y}\phi(y)dy$$

where $f(x)$ is a general function of x.

Let us note that the integral is a convolution

$$\phi(x) - \lambda e^x * \phi(x) = f(x)$$

Now, taking the Laplace transform yields

$$\begin{aligned} \bar{\phi} - \lambda\frac{\bar{\phi}}{s-1} &= \bar{f} \\ \Rightarrow \bar{\phi}\left(\frac{s-1-\lambda}{s-1}\right) &= \bar{f} \\ \Rightarrow \bar{\phi} = \frac{s-1}{s-(1+\lambda)}\bar{f} &= \bar{f} + \frac{\lambda\bar{f}}{s-(1+\lambda)} \\ \Rightarrow \phi = f + \lambda f * e^{(1+\lambda)x} &= f + \lambda\int_0^x f(y)e^{(1+\lambda)(x-y)}dy \end{aligned}$$

3.4 Exercises

Exercise 3.1 Given suitably well behaved functions f, g, and h, establish the following properties of the convolution $f * g$:

(a) $f * g = g * f$

(b) $f * (g * h) = (f * g) * h$

(c) determine f^{-1} such that $f * f^{-1}$, stating any extra properties f must posses in order for the inverse f^{-1} to exist

Work

(a) The proof was given above.

(b) *Proof.*

$$g * h = \varphi(t) = \int_0^t g(\tau)h(t-\tau)h\tau$$

$$f * (g * h) = \int_0^t f(\alpha)\varphi(t-\alpha)d\alpha = \int_0^t \int_0^{t-\alpha} f(\alpha)g(\tau)h(t-\tau-\alpha)d\alpha d\tau$$

Now let us make the substitution $u = t - \tau - \alpha$ and $x = \alpha$ so now

$$dudx = \frac{\partial(u,x)}{\partial(\tau,\alpha)}d\tau d\alpha = \det\begin{bmatrix} -1 & -1 \\ 0 & 1 \end{bmatrix} d\tau d\alpha = -d\tau d\alpha$$

$$\begin{aligned}
\int_0^t \int_0^{t-\alpha} f(\alpha)g(\tau)h(t-\tau-\alpha)d\alpha d\tau &= \int_0^t \int_0^{t-u} f(x)g(t-u-x)h(u)dxdu \\
&= \int_0^t h(u)\left[\int_0^{t-u} g((t-u)-x)f(x)dx\right]du \\
&= \int_0^t h(x)(f*g)(x)dx \\
&= h * (f * g) = (f * g) * h
\end{aligned}$$

□

(c)

$$\begin{aligned}
f * f^{-1} &= 1 \\
\bar{f}\bar{f}^{-1} &= \frac{1}{s} \\
\bar{f}^{-1} &= \frac{1}{s\bar{f}} \\
f^{-1} &= 1 * \mathscr{L}^{-1}\left\{\frac{1}{\bar{f}(s)}\right\} \\
&= \int_0^t \mathscr{L}^{-1}\left\{\frac{1}{\bar{f}(s)}\right\}d\tau
\end{aligned}$$

Exercise 3.2 Use the convolution theorem to establish

$$\mathscr{L}^{-1}\left\{\frac{\bar{f}(s)}{s}\right\} = \int_0^t f(\tau)d\tau$$

Work

$$\begin{aligned}
\mathscr{L}^{-1}\left\{\frac{\bar{f}(s)}{s}\right\} &= \mathscr{L}^{-1}\left\{\bar{f}(s)\right\} * \mathscr{L}^{-1}\left\{\frac{1}{s}\right\} \\
&= f(t) * 1 \\
&= \int_0^t f(\tau)d\tau
\end{aligned}$$

Exercise 3.3 Find the following convolutions

(a) $t * \cos t$

(b) $t * t$

(c) $\sin t * \sin t$

(d) $e^t * t$

(e) $e^t * \cos t$

Work

(a) $t * \cos t = \displaystyle\int_0^t \tau \cos(t-\tau) d\tau = 1 - \cos t$

(b) $t * t = \displaystyle\int_0^t \tau(t-\tau) d\tau = \frac{1}{2} t\tau^2 - \frac{1}{3}\tau^3 \Big|_0^t = \frac{1}{2}t^3 - \frac{1}{3}t^3 = \frac{1}{6}t^3$

(c) $\sin t * \sin t = \displaystyle\int_0^t \sin\tau \sin(t-\tau) d\tau = \tfrac{1}{2}(\sin t - t\cos t)$

(d) $e^t * t = \displaystyle\int_0^t \tau e^{t-\tau} d\tau = e^t \int_0^t \tau e^{-\tau} d\tau = e^t - t - 1$

(e) $e^t * \cos t = \displaystyle\int_0^t e^{t-\tau} \cos\tau d\tau = \tfrac{1}{2}\left(\sin t - \cos t + e^t\right)$

Exercise 3.4

(a) Show that

$$\lim_{x\to 0} \frac{\operatorname{erf}(x)}{x} = \frac{2}{\sqrt{\pi}}$$

(b) Show that

$$\mathscr{L}\left\{t^{-1/2} \operatorname{erf}\sqrt{t}\right\} = \frac{2}{\sqrt{\pi s}} \tan^{-1}\left(\frac{1}{\sqrt{s}}\right)$$

Work

(a)

$$\lim_{x\to 0} \frac{\operatorname{erf}(x)}{x} = \lim_{x\to 0} \frac{\frac{2}{\sqrt{\pi}} e^{-x^2}}{1} = \frac{2}{\sqrt{\pi}}$$

$$\Rightarrow \lim_{s\to\infty} s\mathscr{L}\left\{\frac{\operatorname{erf}(x)}{x}\right\} = \frac{2}{\sqrt{\pi}} \text{ by the initial value theorem}$$

(b) Let us perform a power series expansion of the error function

$$\int_0^{\sqrt{t}} e^{-x^2} dx = \sum_{n=0}^{\infty} (-1)^n \frac{t^{n+1/2}}{(2n+1)\cdot n!}$$

$$t^{-1/2} \operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} \sum_{n=0}^{\infty} (-1)^n \frac{t^n}{(2n+1)\cdot n!}$$

$$\mathscr{L}\left\{t^{-1/2} \operatorname{erf}(x)\right\} = \frac{2}{\sqrt{\pi}} \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)s^{n+1}}$$

$$= \frac{2}{\sqrt{\pi s}} \tan^{-1}\left(\frac{1}{\sqrt{s}}\right)$$

Exercise 3.5 Solve the following differential equations by using Laplace transforms:

(a)
$$\frac{dx}{dt} + 3x = e^{2t} \qquad x(0) = 1$$

(b)
$$\frac{dx}{dt} + 3x = \sin t \qquad x(\pi) = 1$$

(c)
$$\frac{d^2x}{dt^2} + 4\frac{dx}{dt} + 5x = 8\sin t \qquad x(0) = x'(0) = 0$$

(d)
$$\frac{d^2x}{dt^2} - 3\frac{dx}{dt} - 2x = 6 \qquad x(0) = x'(0) = 1$$

(e)
$$\frac{d^2y}{dt^2} + y = 3\sin(2t) \qquad y(0) = 3 \qquad y'(0) = 1$$

Work

(a)
$$\begin{aligned} \frac{dx}{dt} + 3x &= e^{2t} \\ \Rightarrow s\bar{x}(s) - x(0) + 3\bar{x}(s) &= \frac{1}{s-2} \\ \Rightarrow \bar{x}(s) = \frac{1}{(s+3)(s-2)} + \frac{1}{s+3} &= \frac{4}{5}\frac{1}{s+3} + \frac{1}{5}\frac{1}{s-2} \\ \Rightarrow x(t) &= \frac{4}{5}e^{-3t} + \frac{1}{5}e^{2t} \end{aligned}$$

(b)
$$\begin{aligned} \frac{dx}{dt} + 3x &= \sin t \\ \Rightarrow s\bar{x}(s) - x(0) + 3\bar{x}(s) &= \frac{1}{s^2+1} \\ \Rightarrow \bar{x}(s) &= \frac{1}{(s^2+1)(s+3)} + \frac{x(0)}{s+3} \\ &= -\frac{1}{10}\left(\left(\frac{s}{s^2+1} - \frac{3}{s^2+1}\right) - \frac{x(0)}{s+3}\right) \\ \Rightarrow x(t) &= -\frac{1}{10}(\cos t - 3\sin t - x(0)e^{-3t}) \\ \Rightarrow x(\pi) = 1 &= -\frac{1}{10}(-1 - x(0)e^{-3\pi}) \\ \Rightarrow x(0) &= 9e^{3\pi} \\ \Rightarrow x(t) &= \frac{3}{10}\sin t - \frac{1}{10}\cos t + \frac{9}{10}e^{3(\pi-t)} \end{aligned}$$

(c)

$$\begin{aligned}
\frac{d^2x}{dt^2} + 4\frac{dx}{dt} + 5x &= 8\sin t \\
\Rightarrow s^2\bar{x}(s) + 4s\bar{x}(s) + 5\bar{x}(s) &= \frac{8}{s^2+1} \\
\Rightarrow \bar{x}(s) &= \frac{8}{(s^2+1)((s+2)^2+1)} \\
&= \frac{s+2}{(s+2)^2+1} + \frac{1}{(s+2)^2+1} - \frac{s}{s^2+1} + \frac{1}{s^2+1} \\
\Rightarrow x(t) &= e^{-2t}(\cos t - \sin t) - \cos t + \sin t
\end{aligned}$$

(d)

$$\begin{aligned}
\frac{d^2x}{dt^2} - 3\frac{dx}{dt} - 2x &= 6 \\
\Rightarrow s^2\bar{x}(s) - s - 1 - 3s\bar{x} + 3 - 2\bar{x} &= \frac{6}{s} \\
\Rightarrow \bar{x}(s) &= \frac{4s-11}{s^2-3s-2} - \frac{3}{s}
\end{aligned}$$

Let us now complete the square

$$\begin{aligned}
s^2 - 3s - 2 = s^2 - 3s - 2 + \frac{17}{4} - \frac{17}{4} &= \left(s - \frac{3}{2}\right)^2 - \frac{17}{4} \\
4s - 11 = 4\left(s - \frac{11}{4}\right) &= 4\left(s - \frac{6}{4} - \frac{5}{4}\right)
\end{aligned}$$

$$\bar{x}(s) = \frac{4\left(s-\frac{3}{2}\right)}{\left(s-\frac{3}{2}\right)^2 - \frac{17}{4}} - 5\frac{\frac{\sqrt{17}}{2}}{\frac{\sqrt{17}}{2}\left(\left(s-\frac{3}{2}\right)^2 - \frac{17}{4}\right)} - \frac{3}{s}$$

$$x(t) = 4e^{3t/2}\cosh\left(\frac{\sqrt{17}}{2}t\right) - \frac{10}{\sqrt{17}}\sinh e^{3t/2}\left(\frac{\sqrt{17}}{2}t\right) - 3$$

(e)

$$\begin{aligned}
\frac{d^2y}{dt^2} + y &= 3\sin(2t) \\
\Rightarrow \bar{y}(s)(s^2+1) - 3s - 1 &= \frac{6}{s^2+4} \\
\Rightarrow \bar{y}(s) &= -\frac{2}{s^2+4} + \frac{2}{s^2+1} + \frac{3s}{s^2+1} + \frac{3}{s^2+1} \\
\Rightarrow y(t) &= -\sin 2t + 2\sin t + 3\cos t + 3\sin t \\
&= 3\cos t + 5\sin t - \sin 2t
\end{aligned}$$

Exercise 3.6 Solve the following pairs of simultaneous differential equations by using Laplace transforms:

(a)

$$\begin{aligned}
\frac{dx}{dt} - 2x - \frac{dy}{dt} - y &= 6e^{3t} \\
2\frac{dx}{dt} - 3x + \frac{dy}{dt} - 3y &= 6e^{3t} \qquad x(0) = 3 \qquad y(0) = 0
\end{aligned}$$

(b)

$$\begin{aligned}
4\frac{dx}{dt} + 6x + y &= 2\sin(2t) \\
\frac{d^2x}{dt^2} + x - \frac{dy}{dt} &= 3e^{-2t} \qquad x(0) = 2 \qquad x'(0) = -2
\end{aligned}$$

(c)

$$\frac{d^2x}{dt^2} - x + 5\frac{dy}{dt} = t$$
$$\frac{d^2y}{dt^2} - 4y - 2\frac{dx}{dt} = -2 \qquad x(0) = x'(0) = y'(0) = 0 \qquad y(0) = 1$$

Work

(a)

$$\frac{dx}{dt} - 2x - \frac{dy}{dt} - y = 6e^{3t}$$
$$2\frac{dx}{dt} - 3x + \frac{dy}{dt} - 3y = 6e^{3t}$$

$$\begin{cases} (s-2)\bar{x}(s) - (s+1)\bar{y}(s) = \dfrac{6}{s-3} + 3 \\ (2s-3)\bar{x}(s) + (s-3)\bar{y}(s) = \dfrac{6}{s-3} + 6 \end{cases}$$

$$\Rightarrow \bar{x}(s) = \frac{4}{(s-3)(s-1)} + \frac{3s-1}{(s-1)^2} = \frac{2}{s-3} + \frac{1}{s-1} + \frac{2}{(s-1)^2}$$
$$\Rightarrow x(t) = 2e^{3t} + e^t + 2te^t$$

Eliminating $\frac{dy}{dt}$ and solving for y in the original system gives

$$y(t) = -3e^{3t} - \frac{5}{4}x(t) + \frac{3}{4}\frac{dx}{dt}$$
$$= -e^{3t} + e^t - te^t$$

(b) Adding the two equations together gets

$$5\frac{d^2x}{dt^2} + 6\frac{dx}{dt} + x = 4\sin(2t) + 3e^{-2t}$$
$$\Rightarrow \bar{x}(s) = \frac{29}{20(s+1)} + \frac{1}{3(s+2)} + \frac{2225}{1212(5s+1)} - 4\frac{19s-25}{(5s+1)(s+1)}$$
$$\Rightarrow x(t) = \frac{1}{3}e^{-2t} + \frac{29}{20}e^{-t} + \frac{445}{1212}e^{-t/5} - \frac{76}{5-5}\cos(2t) + \frac{48}{505}\sin(2t)$$

Subbing this back into the first ODE gives

$$y(t) = \frac{2}{3}e^{2t} - \frac{29}{10}e^{-t} - \frac{1157}{606}e^{-t/5} + \frac{72}{505}\cos(2t) + \frac{118}{505}\sin(2t)$$

(c) I guess I skipped doing this one after being traumatized by the last problem.

Exercise 3.7 Demonstrate the phenomenon of resonance by solving the equation

$$\frac{d^2x}{dt^2} + k^2x = A\sin(t) \qquad x(0) = x_0 \qquad x'(0) = v_0$$

and showing that the solution is unbounded as $t \to \infty$ for particular values of k.

Work

$$\bar{x}(s^2 + k^2) - sx_0 - v_0 = \frac{A}{s^2+1}$$
$$\bar{x}(s) = \frac{A}{(s^2+1)(s^2+k^2)} + \frac{sx_0}{s^2+k^2} + \frac{v_0}{s^2+k^2}$$

If $k \neq 1$, we can split the first term via partial fractions and invert to get

$$x(t) = \frac{A}{k^2-1}\left(\sin t - \frac{\sin kt}{k}\right) + \frac{v_0}{k}\sin(kt) - x_0\cos(kt)$$

If $k = 1$, we then get a $(1 + s^2)^2$ in the denominator and we therefore convolute $\sin t$ with itself to get

$$x(t) = \frac{A}{2}(\sin t - t\cos t) + v_0 \sin t + x(0)\cos t$$

Where we can see that there is a secular term $t\cos t$ that goes to infinity as $t \to \infty$.

Exercise 3.8 Assuming that the air resistance is proportional to speed, the motion of a particle in air is governed by the equation

$$\frac{w}{g}\frac{d^2x}{dt^2} + b\frac{dx}{dt} = w$$

If the initial conditions at $t = 0$ are $x(0) = 0$ and $v = v_0$, show that the solution is

$$x = \frac{gt}{a} + \frac{(av_0 - g)(1 - e^{-at})}{a^2}$$

where $a = \frac{bg}{w}$. Deduce the terminal speed of the particle.

Work

$$\begin{aligned}
\frac{w}{g}(s^2\bar{x} - v_0) + b(s\bar{x}) &= \frac{w}{s} \\
\bar{x}\left(\frac{w}{g}s^2 + bs\right) &= \frac{w}{s} + \frac{v_0 w}{g} \\
&= \frac{wg + v_0 ws}{sg} \\
\Rightarrow \bar{x} &= \frac{wg + v_0 ws}{ws^3 + gbs^2} \\
&= \frac{g + v_0 s}{s^3 + \frac{bg}{w}s^2} \\
&= \frac{g + v_0 s}{s^2(s + a)} \\
&= \frac{-\frac{1}{a^2}(av_0 - g)}{s + a} + \frac{-\frac{1}{a^2}(av_0 - g)s + \frac{g}{a}}{s^2} \\
\Rightarrow x(t) &= -\frac{1}{a^2}(av_0 - g)e^{-at} - \frac{1}{a^2}(av_0 - g) + \frac{g}{a}t \\
\Rightarrow \frac{dx}{dt} &= \frac{g}{a} + \frac{(av_0 - g)}{a}e^{-at} \\
\Rightarrow \frac{d^2x}{dt^2} &= (g - av_0)e^{-at}
\end{aligned}$$

To find the terminal velocity, set the acceleration to be zero (since at terminal velocity, the velocity isn't changing) and solve for t and plug it back into the expression for the velocity.

Exercise 3.9 Determine the algebraic system of equations obeyed by the transformed electrical variables $\bar{j}_1, \bar{j}_2, \bar{j}_3$ and $\bar{q}_3$ given the electrical circuit equations

$$\begin{aligned}
j_1 &= j_2 + j_3 \\
R_1 j_1 + L_2\frac{dj_2}{dt} &= E\sin(\omega t) \\
R_1 j_1 + R_3 j_3 + \frac{1}{C}q_3 &= E\sin(\omega t) \\
j_3 &= \frac{dq_3}{dt}
\end{aligned}$$

Work

$$\begin{aligned}
\bar{j}_1 - \bar{j}_2 - \bar{j}_3 &= 0 \\
R_1\bar{j}_1 + L_2(s\bar{j}_2 - j_2(0)) &= E\frac{\omega^2}{s^2 + \omega^2} \\
R_1\bar{j}_1 + R_3\bar{j}_3 + \frac{1}{C}\bar{q}_3 &= E\frac{\omega^2}{s^2 + \omega^2} \\
\bar{j}_3 - s\bar{q}_3 &= -q_3(0)
\end{aligned}
\Rightarrow
\begin{bmatrix} 1 & -1 & -1 & 0 \\ R_1 & sL_2 & 0 & 0 \\ R_1 & 0 & R_3 & \frac{1}{C} \\ 0 & 0 & 1 & -s \end{bmatrix}
\begin{bmatrix} \bar{j}_1 \\ \bar{j}_2 \\ \bar{j}_3 \\ \bar{q}_3 \end{bmatrix}
=
\begin{bmatrix} 0 \\ E\frac{\omega^2}{s^2+\omega^2} + j_2(0) \\ E\frac{\omega^2}{s^2+\omega^2} \\ -q_3(0) \end{bmatrix}$$

T

Exercise 3.10 Solve the loaded beam problem expressed by the equation

$$k\frac{dy^4}{dx^4} = \frac{w_0}{c}[c - x + (x-c)H(x-c)] \qquad 0 < x < 2c \qquad y(0) = y'(0) = y''(2c) = y'''(2c) = 0$$

Find the bending moment ky'' at the point $x = \frac{1}{2}c$.

Work

$$\begin{aligned}
s^4\bar{y}(s) &= \frac{w_0}{ck}\left(\frac{c}{s} - \frac{1}{s^2}(1-e^{-cs})\right) + sy''(0) + y'''(0) \\
\Rightarrow \bar{y}(s) &= \frac{w_0}{ck}\left(\frac{c}{s^5} - \frac{1}{s^6}(1-e^{-cs})\right) + \frac{y''(0)}{s^3} + \frac{y'''(0)}{s^4} \\
\Rightarrow y(x) &= \frac{w_0}{ck}\left(\frac{c}{24}x^4 - \frac{1}{120}x^5 + \frac{(x-c)^5}{120}H(x-c)\right) + \frac{1}{2}y''(0)x^2 + \frac{1}{6}y'''(0)x^3 \\
\Rightarrow y''(x) &= \frac{w_0}{ck}\left(\frac{c}{2}x^2 - \frac{1}{6}x^3 + \frac{(x-c)^3}{6}H(x-c)\right) + y''(0) + y'''(0)x \\
\Rightarrow y'''(x) &= \frac{w_0}{ck}\left(cx - \frac{1}{2}x^2 + \frac{(x-c)^2}{2}H(x-c)\right) + y'''(0) \\
\Rightarrow y'''(2c) = 0 &= \frac{w_0}{ck}\left(2c^2 - 2c^2 + \frac{1}{2}c^2\right) + y'''(0) \\
\Rightarrow y'''(0) &= -\frac{w_0 c}{2k} \\
\Rightarrow y''(2c) = 0 &= \frac{w_0}{ck}\left(\frac{c}{2}4c^2 - \frac{1}{3}4c^3 + \frac{1}{6}c^3\right) + y''(0) - \frac{w_0c^2}{k} \\
&= \frac{w_0}{k}\left(2c^2 - \frac{4}{3}c^2 + \frac{1}{6}c^2\right) + y''(0) - \frac{w_0c^2}{k} \\
&= \frac{w_0}{k}\frac{5}{6}c^2 + y''(0) - \frac{w_0c^2}{k} \\
\Rightarrow y''(0) &= \frac{w_0c^2}{6k} \\
\Rightarrow y(x) &= \frac{w_0}{ck}\left(\frac{c}{24}x^4 - \frac{1}{120}x^5 + \frac{(x-c)^5}{120}H(x-c)\right) + \frac{w_0c^2}{12k}x^2 - \frac{w_0c}{12k}x^3
\end{aligned}$$

Exercise 3.11 Solve the integral equation

$$\phi(t) = t^2 + \int_0^t \phi(y)\sin(t-y)dy$$

Work

$$\begin{aligned}
\phi(t) &= t^2 + \phi * \sin t \\
\Rightarrow \bar{\phi}(s) &= \frac{2}{s^3} + \frac{\bar{\phi}}{s^2+1} \\
\Rightarrow \bar{\phi}(s)\left(1 - \frac{1}{s^2+1}\right) &= \bar{\phi}(s)\left(\frac{s^2}{s^2+1}\right) = \frac{2}{s^3} \\
\Rightarrow \bar{\phi}(s) &= \frac{2s^2+2}{s^5} = \frac{2}{s^5} + \frac{2}{s^3} \\
\Rightarrow \bar{\phi}(t) &= \frac{1}{12}t^4 + t^2
\end{aligned}$$

Chapter 4

Fourier Series

4.1 Definition of the Fourier Series

Theorem 4.1 (Bessel's Inequality) *If*

$$\{\mathbf{e_1}, \mathbf{e_2}, \ldots, \mathbf{e_n}, \ldots\}$$

is an orthonormal basis for the vector space V, then for every $\mathbf{a} \in V$ the series

$$\sum_{r=1}^{\infty} |\langle \mathbf{a}, \mathbf{e_n} \rangle|^2$$

converges and the following inequality is true

$$\sum_{r=1}^{\infty} |\langle \mathbf{a}, \mathbf{e_n} \rangle|^2 \leq ||\mathbf{a}||^2$$

Lemma 4.2 (Riemann-Lebesgue) *Let $\{\mathbf{e_1}, \mathbf{e_2}, \ldots\}$ be an orthonormal basis of infinite dimension for the inner product space V. Then for any $\mathbf{a} \in V$,*

$$\lim_{n \to \infty} \langle \mathbf{a}, \mathbf{e_n} \rangle = 0$$

Intuitively, these results can be thought of as for an infinite dimensional space (I believe more specifically a Hilbert Space), the "energy" in each coefficient for every projection onto a basis vector must go to zero for a finite vector. Basically, a finite normed vector "runs out" of stuff to project onto. Let us also note the definitions of pointwise and uniform convergence.

Definition 4.3 (Pointwise Convergence) Let

$$\{f_0, f_1, \ldots, f_m, \ldots\}$$

be a sequence of functinos defined on the closed interval $[a, b]$. We say that the sequence $\{f_0, f_1, \ldots, f_m, \ldots\}$ converges pointwise to f on $[a, b]$ if for each $x \in [a, b]$ and $\epsilon > 0$, there exists a natural number $N(\epsilon, x)$ such that

$$|f_m(x) - f(x)| < \epsilon \ \forall m \geq N(\epsilon, x)$$

Definition 4.4 (Uniform Convergence) Let

$$\{f_0, f_1, \ldots, f_m, \ldots\}$$

be a sequence of functions defined on the closed interval $[a, b]$. We say that the sequence $\{f_0, f_1, \ldots, f_m, \ldots\}$ converges uniformly to f on $[a, b]$ if for each $\epsilon > 0$ there exists a natural number $N(\epsilon)$ such that

$$|f_m(x) - f(x)| < \epsilon \ \forall m \geq N(\epsilon) \text{ and } \forall x \in [a, b]$$

The Fourier series converges pointwise and not uniformly to the function. At discontinuities, it converges to the average of the limit of both sides.

Theorem 4.5 (Dirichlet's Theorem) *If f is a member of the space of piecewise continuous functions which are 2π periodic on the closed interval $[-\pi, \pi]$ and which has both left and right derivatives at each $x \in [-\pi, \pi]$, then for each $x \in [-\pi, \pi]$, the Fourier series of f converges to the value*

$$\frac{f(x^-) + f(x^+)}{2}$$

and at the end points $x = \pm\pi$, the series converges to

$$\frac{f(\pi^-) + f((-\pi)^+)}{2}$$

Definition 4.6 If an interval $[s, t_0]$ can be partitioned into a finite number of subintervals $[s, t_1], [t_1, t_2], [t_2, t_3], \ldots, [t_n, t_0]$ with $s, t_1, t_2, \ldots, t_n, t_0$ an increasing sequence of intervals numbers and such that a given function $f(t)$ is continuous in each of these subintervals but not necessarily at the end points, then $f(t)$ is piecewise continuous in the interval $[s, t_0]$.

Theorem 4.7 *The sequence of functions*

$$\left\{ \frac{1}{\sqrt{2}}, \sin x, \cos x, \sin 2x, \cos 2x, \ldots \right\}$$

form an infinite orthonormal sequence in the space of al piecewise continuous functions on the interval $[-\pi, \pi]$ where the inner product $\langle f, g \rangle$ is defined by

$$\langle f, g \rangle = \frac{1}{\pi} \int_{-\pi}^{\pi} f \bar{g} dx$$

where the overbar denotes the complex conjugate.

The proof of this will be omitted since it is fairly straightforward but tedious. Since this sequence of functions forms an infinite orthonormal basis for the space of piecewise continuous 2π-periodic functions, we can express a piecewise continuous 2π-periodic functions as a linear combination of the basis vectors or

$$f(x) \sim \frac{a_0}{\sqrt{2}} + \sum_{n=1}^{\infty} (a_n \cos(nx) + b_n \sin(nx)) \qquad -\pi < x < \pi$$

Each coefficient represents how much of each basis vector contributes to the overall function and so we can figure this out by projecting our function onto each basis vector. We do this by taking the inner product such that

$$a_0 = \langle f, 1 \rangle = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx$$

$$a_n = \langle f, \cos nx \rangle = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos(nx) dx$$

$$b_n = \langle f, \sin nx \rangle = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin(nx) dx$$

Realistically speaking, we mainly deal with the Fourier expansion as

$$f(x) \sim \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos(nx) + b_n \sin(nx))$$

since the square root in the denominator is annoying to deal with. This technically means that the basis is no longer orthonormal but it is much more convenient to work with. We can also find the Fourier series of other functions by manipulating the series of one that we know. For example, the Fourier series of $y = x$ is

$$y \sim 2 \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} \sin(nx)$$

and so therefore the Fourier series of a generalized straight line $y = mx + c$ is

$$y \sim c + 2m \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} \sin(nx)$$

One notes that since a straight line is an odd function (unless completely horizontal), the Fourier series only consists of odd sin functions.

Definition 4.8 (Even Function) A function $f(x)$ is termed even with respect to the value a is

$$f(a+x) = f(a-x) \ \forall x \in \mathbb{R}$$

Definition 4.9 (Odd Function) A function $f(x)$ is termed odd with respect to the value a if

$$f(a+x) = -f(a-x) \ \forall x \in \mathbb{R}$$

We can then recall Euler's formula and therefore express sin and cos as

$$\cos(nx) = \frac{1}{2}(e^{inx} + e^{-inx})$$
$$\sin(nx) = \frac{1}{2}(e^{inx} - e^{-inx})$$

and so therefore we can express the Fourier series in its complex form as

$$f(x) \sim \sum_{n=-\infty}^{\infty} c_n e^{inx}$$

where

$$c_n = \frac{1}{2}(a_n - ib_n) \qquad c_{-n} = \frac{1}{2}(a_n + ib_n) \qquad c_0 = \frac{1}{2}a_0 \qquad n \in \mathbb{N}$$

We can also write the coefficients in inner product form as

$$c_n = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x)e^{-inx}dx \qquad c_{-n} = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x)e^{inx}dx$$

Let us also consider the half-range series which are defined on the the interval $[0, \pi]$. Our previous basis would not work on this interval but the individual bases

$$\{\sin x, \sin 2x, \sin 3x, \ldots\}$$
$$\left\{\frac{1}{\sqrt{2}}, \cos x, \cos 2x, \ldots\right\}$$

do work. As such, we can effectively have an odd and even extension of a function by taking the half range Fourier series. We therefore have the coefficients to be

$$a_n = \frac{2}{\pi}\int_0^{\pi} f(x)\cos(nx)dx$$

and

$$b_n = \frac{2}{\pi}\int_0^{\pi} f(x)\sin(nx)dx$$

We can also generalize the Fourier series to any interval $[-L, L]$ using the transformation $x \to \frac{n\pi}{L}x$ where the coefficients can be found using the inner product

$$\langle f, g\rangle = \frac{1}{L}\int_{-L}^{L} f\bar{g}dx$$

4.2 Properties of the Fourier Series

Differentiation poses a larger problem than integration of the Fourier series. The reasoning for this is differentiation pulls out an n as a coefficient which for larger values of n, cause the terms to be larger in magnitude than the originals. Integration pulls out a $\frac{1}{n}$ coefficient and therefore the magnitude is smaller.

Theorem 4.10 *If f is continuous on $[-\pi, \pi]$ and piecewise differentiable in $(-\pi, \pi)$ which means that the derivative f' is piecewise continuous on $[-\pi, \pi]$, and if $f(x)$ has the Fourier series*

$$f(x) \sim \frac{a_0}{2} + \sum_{n=1}^{\infty}[a_n\cos(nx) + b_n\sin(nx)]$$

then the Fourier series of the derivative of $f(x)$ is given by

$$f'(x) \sim \sum_{n=1}^{\infty}[-na_n\sin(x) + nb_n\cos(nx)]$$

Note that for differentiating the Fourier series, we need the requirement that the series converges uniformly. Otherwise, differentiating will yield nonsense. The same requirement is not needed for integration. For integration, we have the following theorem.

Theorem 4.11 *If f is piecewise continuous on the interval $[-\pi, \pi]$ and has the Fourier series*

$$f(x) \sim \frac{a_0}{2} + \sum_{n=1}^{\infty}[a_n \cos(nx) + b_n \sin(nx)]$$

then for each $x \in [-\pi, \pi]$,

$$\int_{-\pi}^{x} f(t)dt = \frac{a_0(x+\pi)}{2} + \sum_{n=1}^{\infty}\left[\frac{a_n}{n}\sin(nx) - \frac{b_n}{n}(\cos(nx) - \cos(n\pi))\right]$$

We can prove the more general result. Let $F(t)$ be represented as

$$F(t) \sim \frac{A_0}{2} + \sum_{n=1}^{\infty}(A_n \cos(nt) + B_n \sin(nt))$$

along with the usual periodicity requirements. Let us define another function $G(x)$ such that

$$\int_{-\pi}^{t} G(x)dx = \frac{1}{2}a_0 t + F(t)$$

Each of the Fourier coefficients are defined under the usual inner product as $A_n = \langle F(t), \cos(nt)\rangle$ and $B_n = \langle F(t), \sin(nt)\rangle$. Using the Fundamental Theorem of Calculus, we have that $F'(t) = G(t) - \frac{1}{2}a_0$. If we integrate by parts once for A_n and B_n, we have

$$A_n = \frac{1}{\pi}\int_{-\pi}^{\pi} F(t)\cos(nt)dt = -\frac{1}{n\pi}\int_{-\pi}^{\pi} F'(t)\sin(nt)dt$$
$$B_n = \frac{1}{\pi}\int_{-\pi}^{\pi} F(t)\sin(nt)dt = -\frac{1}{n\pi}\int_{-\pi}^{\pi} F'(t)\cos)nt)dt$$

If we write the Fourier coefficients of $G(t)$ as a_n and b_n, we see that

$$A_n = -\frac{b_n}{n} \qquad B_n = \frac{a_n}{n}$$

We now write the Fourier series of $F(t)$ as

$$F(t) = \frac{A_0}{2} + \sum_{n=1}^{\infty}\left(-\frac{b_n}{n}\cos(nt) + \frac{a_n}{n}\sin(nt)\right)$$

Let us now consider the values of $F(t)$ at the end points $t = \pm\pi$. For $t = \pi$, we have

$$F(\pi) = -\frac{a_0\pi}{2} + \int_{-\pi}^{\pi} G(x)dx$$

and for the other endpoint we have

$$F(-\pi) = \frac{a_0\pi}{2}$$

and by the periodicity requirement, $F(\pi) = F(-\pi) = \frac{a_0\pi}{2}$. Assuming there is no jump discontinuity (and therefore piecewise differentiable), we have that

$$\frac{a_0\pi}{2} = \frac{A_0}{2} - \sum_{n=1}^{\infty}\frac{b_n}{n}\cos(n\pi) = \frac{A_0}{2} - \sum_{n=1}^{\infty}(-1)^n\frac{b_n}{n}$$
$$\Rightarrow \frac{A_0}{2} = \frac{a_0\pi}{2} + \sum_{n=1}^{\infty}(-1)^n\frac{b_n}{n}$$

And we can now substitute this into our Fourier series for $F(t)$

$$F(t) \sim \frac{1}{2}a_0\pi + \sum_{n=1}^{\infty}\frac{1}{n}[b_n((-1)^n - \cos(nt)) + a_n\sin(nt)]$$

Using our previous expression for the integral of $G(x)$, we get

$$\int_{-\pi}^{t} G(x)dx = \frac{1}{2}a_0 t + F(t) = \frac{1}{2}a_0(\pi + t) + \sum_{n=1}^{\infty}\frac{1}{n}[b_n(-1)^n - b_n\cos(nt) + a_n\sin(nt)]$$

The function $G(\xi)$ can be defined in a similar way and by subtracting $G(t) - G(\xi)$ and applying properties of definite integrals, we get

$$\int_{\xi}^{t} G(x)dx = \frac{1}{2}a_0(t-\xi) + \sum_{n=1}^{\infty}\frac{1}{n}[b_n(\cos(n\xi) - \cos(nt)) + a_n(\sin(nt) - \sin(n\xi))]$$

Theorem 4.12 *If $f(t)$ and $g(t)$ are continuous in $(-\pi, \pi)$ and square integrable, that is*

$$\int_{-\pi}^{\pi}|f(t)|^2dt \text{ and } \int_{-\pi}^{\pi}|g(t)|^2dt < \infty$$

and a_n, b_n the Fourier coefficients of $f(t)$ and α_n, β_n the Fourier coefficients of $g(t)$, then

$$\int_{-\pi}^{\pi} f(t)g(t)dt = \frac{1}{2}\pi a_0\alpha_0 + \pi\sum_{n=1}^{\infty}(\alpha_n a_n + \beta_n b_n)$$

Proof. We can write the product of these two functions as

$$f(t)g(t) \sim \frac{1}{2}a_0 g(t) + \sum_{n=1}^{\infty}(a_n g(t)\cos(nt) + b_n g(t)\sin(nt))$$

We then integrate both sides from π to $-\pi$ and since it is uniformly convergent (by some magic since it obeys the Cauchy Schwarz inequality and is therefore finite), we can swap the sum and integral operators to get

$$\int_{-\pi}^{\pi} f(t)g(t)dt = \frac{1}{2}a_0\int_{-\pi}^{\pi} g(t)dt + \sum_{n=1}^{\infty}\left[a_n\int_{-\pi}^{\pi} g(t)\cos(nt)dt + b_n\int_{-\pi}^{\pi} g(t)\sin(nt)dt\right]$$

Recognizing that the integrals lead to an inner product between $g(t)$ and our Fourier basis, we have that

$$\int_{-\pi}^{\pi} f(t)g(t)dt = \frac{1}{2}\pi a_0\alpha_0 + \pi\sum_{n=1}^{\infty}(a_n\alpha_n + \beta_n b_n)$$

□

Theorem 4.13 (Parseval's Theorem) *If $f(t)$ is continuous in the range of $(-\pi, \pi)$, is square integrable, and has Fourier coefficients a_n, b_n then*

$$\int_{-\pi}^{\pi}|f(t)|^2dt = 2\pi a_0^2 + \pi\sum_{n=1}^{\infty}(a_n^2 + b_n^2)$$

Proof. Use Theorem 4.12 by setting $g(t) = f(t)$. □

4.3 Generalized Fourier Series

Instead of using just trig functions as a basis, we can generalize the principle further by using a set of functions $\phi_n(t)$ such that they span the function space and are orthogonal to each other. We can therefore express $f(t)$ as

$$f(t) = \sum_{n=1}^{\infty} k_n\phi_n(t)$$

We find the coefficients k_n via the inner product

$$\int_{t_0}^{t_1} f(t)\phi_n(t)dt = \sum_{m=1}^{\infty} k_m \int_{t_0}^{t_1} \phi_m(t)\phi_n(t)$$

where

$$\int_{t_0}^{t_1} \phi_m(t)\phi_n(t) = \delta_{m,n}\gamma_n$$

where $\delta_{m,n}$ is the Kronecker delta defined as

$$\delta_{m,n} = \begin{cases} 0 & m \neq n \\ 1 & m = n \end{cases}$$

Since the functions may not necessarily be orthonormal, we have

$$\gamma_n = \int_{t_0}^{t_1} |\phi_n(t)|^2 dt$$

Therefore γ_n acts as the normalization factor and

$$k_n = \frac{1}{\gamma_n} \int_{t_0}^{t_1} f(t)\phi_n(t)dt$$

We can also state Parseval's theorem for generalized Fourier series

$$\int_{t_0}^{t_1} |f(t)|^2 dt = \sum_{n=1}^{\infty} \gamma k_n^2$$

The basis of this theory comes from Sturm-Liouville Theory which concerns the Sturm-Liouville eigenfunction problem

$$\frac{d}{dx}\left[p(x)\frac{dy}{dx}\right] + q(x)y = -\lambda w(x)y$$

on the finite interval $[a, b]$ that is regular. Regular implies the following conditions:

- the coefficient functions p, q, w and p' are all continuous on $[a, b]$
- $p(x), w(x) > 0 \ \ \forall x \in [a, b]$
- The problem has the Robin boundary conditions

$$\begin{cases} \alpha_1 y(a) + \alpha_2 y'(a) = 0, & \alpha_1, \alpha_2 \neq 0 \\ \beta_1 y(b) + \beta_2 y'(b) = 0, & \beta_1, \beta_2 \neq 0 \end{cases}$$

The theory states the following results:

- There are an infinite number of eigenvalues that can be ordered such that $\lambda_1 < \lambda_2 < \ldots < \lambda_n < \ldots \to \infty$
- Corresponding to each eigenvalue λ_n, there exists a unique eigenfunction $y_n(x)$ deemed the nth *fundamental solution*
- The normalized eigenfunctions $y_n(x)$ form an orthonormal basis under the w-weighted inner product in the Hilbert Space $L^2([a, b], w(x)dx)$, or in other words

$$\langle y_n, y_m \rangle = \int_a^b y_n(x)y_m(x)w(x)dx = \delta_{n,m}$$

We can typically write the Sturm-Liouville problem in the form

$$L[y(x)] = -\lambda w(x)y(x)$$

where L is a second order differential operator and $w(x)$ is the weighting function. Some examples of Sturm-Liouville problems include:

- The time-independent Schrödinger's Equation (which can be written in the form $H\Psi = E\Psi$ where H is the Hamiltonian operator)

$$\left(-\frac{\hbar^2}{2m}\frac{d}{dx^2} + V(x)\right)\Psi = E\Psi$$

- Bessel's Equation

$$x^2\frac{d^2y}{dx^2} + x\frac{dy}{dx} + (x^2 - \alpha^2)y = 0$$

- Legendre's Equation

$$(1 - x^2)\frac{d^2y}{dx^2} - 2x\frac{dy}{dx} + n(n+1)y = 0$$

Chapter 5

Partial Differential Equations

5.1 Introduction

Partial differential equations (PDEs) that are second order and linear can be classified into three main groups. If we look at a general linear second order PDE, that is

$$Au_{xx} + Bu_{xy} + Cu_{yy} + (\text{first order terms}) = 0$$

We can express the determinant as $D = B^2 - 4AC$ and therefore classify linear 2nd order PDEs into three categories: hyperbolic ($D > 0$), elliptic ($D < 0$), and parabolic ($D = 0$). One example of a hyperbolic equation is the one dimensional wave equation

$$\frac{\partial^2 u}{\partial x^2} = \frac{1}{c^2}\frac{\partial^2 u}{\partial t^2}$$

This equation appears almost everywhere and is widely studied. It can be used to describe the displacement of a string as well as the density changes of a sound wave propagating through a medium. By applying a key identity from vector calculus and a free space condition to Faraday's and Ampere's Law, James Clerk Maxwell was able to calculate the speed of light to be $c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}}$ and showed that electricity and magnetism were the same phenomena.

An example of an elliptic PDE is Laplace's equation which is arguably the most important PDE in physics.

$$\nabla^2 u = 0$$

where $\nabla^2 = \Delta = \sum_{i=1}^{n} \frac{\partial^2}{\partial x_i^2}$ denotes the Laplacian operator in $\mathbb{R}^n$. Solutions to Laplace's equation are known as harmonics. Harmonic functions in a domain $D \in \mathbb{R}^n$ have maximums on the border of D, ∂D, as opposed to in D. Harmonic functions are extremely well-behaved. So much so that one can find the value of a point inside a closed path γ by knowing the values of the function and its derivative along γ. For real valued functions, this is done via Green's Representation and Poisson Integral formula. For holomorphic functions (since they are also harmonic as they abide by the Cauchy-Riemann equations), we use Cauchy Integral Formula.

The final example is one of a parabolic equation: the one dimensional heat equation.

$$\frac{\partial u}{\partial t} = \kappa \frac{\partial^2 u}{\partial x^2}$$

This equation is actually the genesis of the Fourier series. Fourier was studying the solutions to the heat equation on a ring and due to the periodicity requirement, required for the solutions to be comprised of sin and cos functions. Since the initial conditions could be anything, Fourier reasoned any function could be represented as an infinite sum of trigonometric functions. Olver's introduction to Fourier Series in his book *Introduction to Partial Differential Equations* is perhaps the best introduction to Fourier series I have read and is just so good as he goes through how Fourier was able to come to his conclusion.

5.2 Separation of Variables

Perhaps one of the most well used techniques for solving PDEs is separation of variables. Let us take the one dimensional heat equation to demonstrate the idea. Let us assume there is some function $u(t, x)$ that satisfies the

equation

$$\frac{\partial u}{\partial t} = \kappa \frac{\partial^2 u}{\partial x^2}$$

and any associated boundary conditions. Then let us make the ansatz that the solution takes the form

$$u(t,x) = T(t)X(x)$$

The heat equation then becomes

$$T'X = \kappa T X''$$

We then divide both sides by $u = TX$ to get

$$\frac{T'}{T} = \kappa \frac{X''}{X}$$

Since both sides of the equation are independent of each other, they must equal the same constant. In this case, let this constant be $-\alpha^2$ such that we now have the ODEs

$$T' = -\alpha^2 T$$

$$X'' = -\frac{\alpha^2}{\kappa} X$$

These are well known differential equations and give the general solutions

$$T(t) = Ae^{-\alpha^2 t} \qquad X(x) = B\cos\left(\frac{\alpha x}{\sqrt{\kappa}}\right) + C\sin\left(\frac{\alpha x}{\sqrt{\kappa}}\right)$$

If the initial temperature distribution at $t = 0$ is $u(0,x) = f(x)$, then we must have that

$$f(x) = B\cos\left(\frac{\alpha x}{\sqrt{\kappa}}\right) + C\sin\left(\frac{\alpha x}{\sqrt{\kappa}}\right)$$

but this is generally not true. To resolve this issue, let us remember that any temperature distribution on the interval $[0, L]$ can be written as the infinite sum of sin and cos functions. Therefore

$$f(x) \sim \sum_{n=1}^{\infty} B_n \cos\left(\frac{\alpha_n x}{\sqrt{\kappa}}\right) + C_n \sin\left(\frac{\alpha_n x}{\sqrt{\kappa}}\right)$$

If we assume homogeneous boundary conditions (i.e. $u(t,0) = u(t,L) = 0$), B_n must be zero for all n and we require

$$\frac{\alpha_n}{\sqrt{\kappa}} = \frac{n\pi}{L}$$

Therefore the general solution is

$$u(t,x) = \sum_{n=1}^{\infty} C_n e^{-\frac{1}{L^2} n^2 \pi^2 \kappa} \sin\left(\frac{n\pi x}{L}\right)$$

where the Fourier coefficients are found via $C_n = \left\langle f(x), \sin\left(\frac{n\pi x}{L}\right)\right\rangle$. One sees that the higher frequency contributions, the frequencies associated with large n, are dampened quickly by the exponential decay term. As such, the heat equation is often used to denoise signals.

5.3 Laplace Transforms for PDEs

The Laplace transform is convenient for PDEs in which the domain aligns nicely with the range of the Laplace transform's bounds. That is from $0 \to \infty$. For most PDEs, this is the time domain. We can also apply the derivative property (2.1) to substitute in our initial conditions nicely. Additionally, assuming that a solution $u \in C^k$ and the integral is well defined, we can swap the differentiation and integration operators to get

$$\mathscr{L}\left\{\frac{\partial^k u(x_1, x_2, \ldots, x_n)}{\partial x_i^k}\right\}_{x_j} = \int_0^\infty \frac{\partial^k}{\partial x_i^k} u e^{-sx_j} dx_j = \frac{\partial^k}{\partial x_i^k} \int_0^\infty u e^{-sx_j} dx_j = \frac{\partial^k}{\partial x_i^k} \bar{u} \qquad i,j \in \{1,\ldots,n\} \qquad i \neq j$$

For the one dimensional heat equation, we can then generally get

$$s\bar{u} - u(0) = \frac{\partial^2 \bar{u}}{\partial x^2}$$

We now have a linear second order ODE which can be solved relatively easily. The next step is to transform any boundary conditions into the s-domain and then solve for the variables then. Afterwards, we invert the function to retrieve the $u(t,x)$ we want.

5.4 Asymptotic Expansions

Oftentimes we cannot find an exact solution. However, we can approximate things to a certain degree. Let us state Watson's Lemma.

Lemma 5.1 (Watson's Lemma) *Let $0 < T \leq \infty$ be fixed. Assume $\varphi(t) = t^\lambda g(t)$ where $g(t)$ is infinitely differentiable around $t = 0$, $g(0) \neq 0$, and $\lambda > -1$. Also suppose that either $g(t)$ is of exponential order or the integral of the norm of $\varphi(t)$ is finite. Then it is true for all positive x that*

$$\left| \int_0^T \varphi(t) e^{-xt} dt \right| < \infty$$

and

$$\int_0^T \varphi(t) e^{-xt} dt = \sum_{n=1}^{\infty} \frac{g^{(n)}(0)\Gamma(\lambda + n + 1)}{n! x^{\lambda+n+1}}$$

Proof. Ask a toddler on the street. □

A special case arises when $T = \infty$ as the left hand side then becomes a Laplace transform and we then have

$$\bar{\varphi}(s) = \sum_{n=1}^{\infty} a_n \frac{\Gamma(n + \lambda + 1)}{s^{\lambda+n+1}}$$

and the value of the expansion

$$\varphi(t) = \sum_{n=1}^{\infty} a_n t^{\lambda+n}$$

can be approximated for small values of t by the Initial Value Theorem (2.7) for finding the value of the transformed series as $s \to \infty$.

Chapter 6

Fourier Transforms

6.1 Derivation of the Fourier Transform

Definition 6.1 Let f be a function defined for all $x \in \mathbb{R}$ with values in $\mathbb{C}$. The Fourier transform is a mapping $f : \mathbb{R} \mapsto \mathbb{C}$ defined by

$$F(\omega) = \int_{-\infty}^{\infty} f(x)e^{-i\omega x}dx$$

Let us relate the Fourier series with the Fourier transform. Let us consider a function $f(t)$. Let us now cconsider a function $g(t)$ that has a periodicity of T and effectively takes the "chunk" of $f(t)$ from the interval $\left[-\frac{T}{2}, \frac{T}{2}\right]$ and repeats it for all t. We also make the transformation $x \to \frac{2\pi t}{T}$. Our function $g(t)$ can be written as

$$g(t) = \begin{cases} f(t) & |t| < \frac{1}{2}T \\ f(t - nT) & \frac{1}{2}(2n-1)T < |t| < \frac{1}{2}(2n+1)T \end{cases}$$

We can then write it as a complex Fourier series

$$g(t) = \sum_{n=-\infty}^{\infty} G_n e^{in\omega_0 t} \qquad G_n = \frac{1}{T}\int_{-T/2}^{T/2} g(t)e^{-in\omega_0 t}dt$$

where $\omega_0 = \frac{2\pi}{T}$ which comes from $x = \frac{2\pi t}{T} = \omega_0 t$. We require the transformation to have 2π and not just π since the complex exponential is only periodic over 2π. We now substitute the Fourier coefficient formula into the Fourier series to get

$$g(t) = \sum_{n=-\infty}^{\infty} \left[\frac{1}{T}\int_{-T/2}^{T/2} g(t)e^{-in\omega_0 t}dt\right] e^{in\omega_0 t}$$

Let us now note that the frequency spacing $\Delta\omega = (n+1)\omega_0 - n\omega_0 = \omega_0$ and rename $n\omega_0 = \omega_n$ We can therefore rewrite $\frac{1}{T} = \frac{\Delta\omega}{2\pi}$ and G_n as

$$G_n = \int_{-T/2}^{T/2} g(t)e^{-i\omega_n t}dt$$

Let us now substitute this back into our Fourier series and take the limit as $T \to \infty$

$$g(t) = \lim_{T\to\infty} \sum_{n=-\infty}^{\infty} G_n e^{i\omega_n t}\frac{\Delta\omega}{2\pi}$$

The frequency spacing becomes an infinitesimal and the sum becomes an integral so we have

$$g(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty}\left[\int_{-\infty}^{\infty} g(t)e^{-i\omega t}dt\right]e^{i\omega t}d\omega$$

We can now see that the inner integral is the Fourier transform of $g(t)$ and

$$\mathscr{F}^{-1}\left\{G(\omega)\right\} = \frac{1}{2\pi}\int_{-\infty}^{\infty} G(\omega)e^{i\omega t}dt$$

is the inverse. In essence, we were effectively taking the Fourier series representation and then letting the period become infinitely wide to "encompass" the entire function $f(t)$ that we wanted the transform of.

Using this, we can give a brief proof of the inverse Laplace transform. Let us consider a complex valued function $F(\omega)$ of the form

$$F(\omega) = A(\omega)e^{i\phi(\omega)}$$

where $A(\omega)$ and $\phi(\omega)$ are real value functions of $\omega \in \mathbb{R}$. $A(\omega)$ acts as the amplitude and $\phi(\omega)$ acts as the phase angle. Using the inverse Fourier transform, we see that

$$f(x) = \frac{1}{2\pi}\int_{-\infty}^{\infty} A(\omega)e^{i\omega x + i\phi(\omega)}d\omega$$

However, this is often not very useful since the oscillatory nature leads to a lot of integrals not converging. As such, let instead do the Fourier transform of the function $e^{-kx}f(x)$ where the exponential decay term allows more functions to converge. We then change the lower bound of integration to 0 and assuming $f(x)$ is of exponential order, we have

$$F_k(\omega) = \int_0^{\infty} e^{-(k+i\omega)x}f(x)dx$$

Finding the inverse of F_k using our definition for the inverse Fourier transform gives us

$$\frac{1}{2\pi}\int_{-\infty}^{\infty} e^{i\omega x}F_k(\omega)d\omega = \begin{cases} 0 & x < 0 \\ e^{-kx}f(x) & x \geq 0 \end{cases}$$

Multiplying both sides to cancel out the damping factor on the right hand side (and bringing it under the integral side on the left), we have

$$\frac{1}{2\pi}\int_{-\infty}^{\infty} e^{(k+i\omega)x}F_k(\omega)d\omega = \begin{cases} 0 & x < 0 \\ f(x) & x \geq 0 \end{cases}$$

We now make the substitution $s = k + i\omega \to ds = id\omega$ so now the limits of integration will be $s = k \pm i\infty$ when $\omega = \pm\infty$ where we now have

$$f(x) = \frac{1}{2\pi i}\int_{k-i\infty}^{k+i\infty} e^{sx}F(s)ds$$

6.2 Properties of the Fourier Transform

As with the Fourier series we can represent the decomposition as a complex exponential or in terms of sin and cos, we can do the same with the Fourier transform. Let us define the Fourier sin and cos transforms to be

$$F_s(\omega) = \int_0^{\infty} f(t)\sin(\omega t)dt$$
$$F_c(\omega) = \int_0^{\infty} f(t)\cos(\omega t)dt$$

We change the bottom limit to 0 for a reason (although other texts may keep the bottom as $-\infty$. We can then write the Fourier transform of $F(\omega)$ as

$$F(\omega)\int_0^{\infty}[f(t) + f(-t)]\cos(\omega t)dt - i\int_0^{\infty}[f(t) - f(-t)]\sin(\omega t)dt$$

We can now see why changing the bottom bound to 0 is nice as if the function is odd, $F(\omega)$ is purely imaginary. If $f(t)$ is even, then $F(\omega)$ is completely real. The inverse of these transforms are

$$f(t) = \frac{2}{\pi}\int_0^{\infty} F_s(\omega)\sin(\omega t)dt$$
$$f(t) = \frac{2}{\pi}\int_0^{\infty} F_c(\omega)\cos(\omega t)dt$$

The Fourier transform also has a derivative property like the Laplace transform which is once again proven by integration by parts and induction which we shall omit here.

$$\mathscr{F}\left\{\frac{d^n f}{dt^n}\right\} = (i\omega)^n F(\omega)$$

Theorem 6.2 (Lebesgue Dominated Convergence Theorem) *Let f_h such that $h \in \mathbb{R}$ be a family of piecewise continuous functions. If*

1. *There exists a function g such that $|f_h(x)| \leq g(x) \ \forall x \in \mathbb{R}$ and all $h \in \mathbb{R}$*
2. $\int_{-\infty}^{\infty} g(x)dx < \infty$
3. $\lim_{h\to 0} f_h((x) = f(x) \ \forall x \in R$

then

$$\lim_{h\to 0} \int_{-\infty}^{\infty} f_h(x)dx = \int_{-\infty}^{\infty} f(x)dx$$

Let us now consider the vector space $G\{\mathbb{R}\}$ which is the family of functions where every $f \in G\{\mathbb{R}\}$ is such that $f : \mathbb{R} \mapsto \mathbb{C}$, is piecewise continuous, and absolutely integrable. These are the family of functions that have a Fourier transform.

Theorem 6.3 *For each $f \in G\{\mathbb{R}\}$,*

1. *$F(\omega)$ is defined for all $\omega \in \mathbb{R}$*
2. *F is a continuous function on $\mathbb{R}$*
3. $\lim_{\omega\to\pm\infty} F(\omega) = 0$

Proof. To prove this, we need to use the Lebesgue Dominated Convergence Theorem. We know the magnitude of the oscillatory kernel $|e^{i\omega x}| = 1$ and since $f(x)$ is absolutely integrable by definition, we have

$$\int_{-\infty}^{\infty} |f(x)e^{-i\omega x}|dx = \int_{-\infty}^{\infty} |f(x)|dx < \infty$$

which proves that $F(\omega)$ is well defined for all ω. We now prove that $F(\omega)$ is continuous by considering

$$\begin{aligned} F(\omega + h) - F(\omega) &= \int_{-\infty}^{\infty} [f(x)e^{-i\omega(x+h)} - f(x)e^{-i\omega x}]dx \\ &= \int_{-\infty}^{\infty} f(x)e^{-i\omega x}[e^{-i\omega h} - 1]dx \end{aligned}$$

By letting $f_h(x) = f(x)e^{-i\omega x}[e^{-i\omega h} - 1]$, we can easily see that $\lim_{h\to 0} f_h(x) = 0 \ \forall x \in \mathbb{R}$. By noting that the norm of $|e^{-i\omega h}| = 1$ and applying the triangle inequality, we have

$$|f_h(x)| = |f(x)||e^{-i\omega x}||e^{-i\omega h} - 1| \leq |f(x)|(|e^{-i\omega h}| + |-1|) \leq 2|f(x)|$$

We can therefore let our function $g(x) = 2|f(x)|$ and since $f(x)$ is absolutely integrable, so is $g(x)$. We have therefore fulfilled all conditions of the Lebesgue Dominated Convergence Theorem. We can now say that

$$\lim_{h\to 0} \int_{-\infty}^{\infty} f_h(x)dx = \int_{-\infty}^{\infty} |f(x)e^{-i\omega x}[0]|dx = 0$$

and hence

$$\lim_{h\to 0} [F(\omega + h) - F(\omega)] = 0$$

which is the condition for continuity. To prove the final statement, consider the Fourier transform of $f(x)$ broken down into its real and imaginary components

$$F(\omega) = \int_{-\infty}^{\infty} f(x)\cos(\omega x)dx - i\int_{-\infty}^{\infty} f(x)\sin(\omega x)dx$$

By recognizing each of these parts are an inner product between $f(x)$ and some basis vector $\cos(\omega x)$ or $\sin(\omega x)$, we can apply the Riemann-Lebesgue Lemma (4.2) and show that

$$\lim_{\omega\to\pm\infty} F(\omega) = \lim_{\omega\to\pm\infty} \int_{-\infty}^{\infty} f(x)\cos(\omega x)dx - \lim_{\omega\to\pm\infty} i\int_{-\infty}^{\infty} f(x)\sin(\omega x)dx = 0$$

□

Theorem 6.4 *Let $f(x) \in G\{\mathbb{R}\}$ and $a, b \in \mathbb{R}$ with $a \neq 0$ where we denote the Fourier transform of f as $\mathscr{F}\{f\} = \hat{f}$. Let $g(x) = f(ax+b)$, then*

$$\mathscr{F}\{g(x)\} = \frac{1}{|a|} e^{i\omega b/a} \hat{f}\left(\frac{\omega}{a}\right)$$

Proof. We start by simply using the definition of the Fourier transform

$$\mathscr{F}\{g(x)\} = \int_{-\infty}^{\infty} f(ax+b) e^{-i\omega x} dx$$

Let us now make the substitution $t = ax + b \to dt = a dx$ where if $a > 0$

$$\mathscr{F}\{g(x)\} = \frac{1}{a} \int_{-\infty}^{\infty} f(t) e^{-i\omega(t-b)/a} dt = \frac{1}{a} e^{i\omega b/a} \int_{-\infty}^{\infty} f(t) e^{-i\omega t/a} dt = \frac{1}{a} e^{i\omega b/a} \hat{f}\left(\frac{\omega}{a}\right)$$

and if $a < 0$

$$\mathscr{F}\{g(x)\} = -\frac{1}{a} e^{i\omega b/a} \hat{f}\left(\frac{\omega}{a}\right)$$

(the negative comes from reswapping the bounds of integration after the substitution) and so thus we have

$$\mathscr{F}\{g(x)\} = \frac{1}{|a|} e^{i\omega b/a} \hat{f}\left(\frac{\omega}{a}\right)$$

□

6.3 Fourier Transforms for PDEs

Fourier transforms are often impractical for many PDE applications due to the limited amount of functions you can actually take the Fourier transform of. It also requires the bounds to be infinite in order to accommodate the bounds of integration in the transform. However, one can use the finite Fourier transform

$$\int_a^b f(t) e^{-i\omega t} dt$$

to work with bounds you want to work on and then make the function doubly periodic. This does raise some convergence issues but I will not go into more detail about them. The Fourier transform, even when it is relevant to a problem, is still oftentimes impractical. However, there are still a few cases such as an infinitely long bar or solving a Laplace's equation in a plane where we can take a multi dimensional Fourier transforms. We shall now prove some convolution identities. We should also note that the general convolution operator is defined as

$$f(t) * g(t) = \int_{-\infty}^{\infty} f(\tau) g(t-\tau) d\tau$$

and the reason for the different with the one for Laplace transforms is that the Laplace use case is for causal signals that have a unit function baked in that leads to the bounds changing to $0 \to t$.

$$\begin{aligned} f * g &= \int_{-\infty}^{\infty} f(\tau) g(t-\tau) d\tau \\ \mathscr{F}\{f * g\} &= \int_{-\infty}^{\infty} e^{-i\omega t} \int_{-\infty}^{\infty} f(\tau) g(t-\tau) d\tau dt \\ &= \int_{-\infty}^{\infty} f(\tau) \int_{-\infty}^{\infty} g(t-\tau_) e^{-ixt} dt d\tau \end{aligned}$$

Now we make the substitution $u = t - \tau \to du = dt$

$$\begin{aligned} \mathscr{F}\{f * g\} &= \int_{-\infty}^{\infty} f(\tau) \int_{-\infty}^{\infty} g(u) e^{-i\omega(u+\tau)} du d\tau \\ &= \int_{-\infty}^{\infty} f(\tau) e^{-i\omega\tau} d\tau \int_{-\infty}^{\infty} g(u) e^{-i\omega u} du \\ &= \hat{f}(\omega) \hat{g}(\omega) \end{aligned}$$

We shall also prove that multiplication in the time domain is equivalent to a convolution in the frequency domain.

$$\begin{aligned}
\mathscr{F}\{f(t)g(t)\} &= \int_{-\infty}^{\infty} f(t)g(t)e^{-ikt}dt \\
&= \int_{-\infty}^{\infty} \left[\frac{1}{2\pi}\int_{-\infty}^{\infty} \hat{f}(\kappa)e^{i\kappa t}d\kappa\right] g(t)e^{-ikt}dt \\
&= \frac{1}{2\pi}\int_{-\infty}^{\infty} \hat{f}(\kappa)\left[\int_{-\infty}^{\infty} g(t)e^{-i(k-\kappa)t}dt\right]d\kappa \\
&= \frac{1}{2\pi}\int_{-\infty}^{\infty} \hat{f}(\kappa)\hat{g}(k-\kappa)d\kappa \\
&= \frac{1}{2\pi}\hat{f}(k) * \hat{g}(k)
\end{aligned}$$

6.4 Windowing

In practice, it is often impossible to take a full Fourier transform due to the limits of integration. However, we note that the Fourier coefficients of the complex Fourier series look suspiciously like a finite Fourier transform. As such, we can obtain the 2π-Fourier transform by finding $2\pi c_n$ and inverting using the complex Fourier series where

$$c_n = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(t)e^{-int}dt$$

In practice, we want to restrict the domain to some window of our choosing. As such, let us find the Fourier transform of the Dirac-δ function

$$\mathscr{F}\{\delta(t-t_0)\} = \int_{-\infty}^{\infty} \delta(t-t_0)e^{-i\omega t}dt = e^{-i\omega t_0}$$

If we want to find the function that leads to a δ function in the frequency domain, let us assum that $F(\omega) = \delta(\omega-\omega_0)$ and then appy the inversion formula

$$\begin{aligned}
f(t) &= \frac{1}{2\pi}\int_{-\infty}^{\infty} \delta(\omega-\omega_0)e^{i\omega t}d\omega = \frac{1}{2\pi}e^{i\omega_0 t} \\
\Rightarrow \mathscr{F}\left\{e^{i\omega_0 t}\right\} &= 2\pi\delta(\omega-\omega_0)
\end{aligned}$$

Assuming the complex Fourier series of $f(t)$ can be written as

$$f(t) = \sum_{n=-\infty}^{\infty} F_n e^{in\omega_0 t}$$

we can find the Fourier transform of $f(t)$ by finding the Fourier transform of each of the individual terms of the Fourier series representation (since the Fourier transform is a linear operator)

$$\begin{aligned}
\mathscr{F}\{f(t)\} &= \mathscr{F}\left\{\sum_{n=-\infty}^{\infty} F_n e^{in\omega_0 t}\right\} \\
&= \sum_{n=-\infty}^{\infty} F_n \mathscr{F}\left\{e^{in\omega_0 t}\right\} \\
&= 2\pi\sum_{n=-\infty}^{\infty} F_n\delta(\omega-\omega_0)
\end{aligned}$$

In the specific case that our function $f(t)$ is the Shah function (since it looks like the Cyrillic letter sha), an infinite train of δ functions,

$$f(t) = \sum_{n=-\infty}^{\infty} \delta(t-nT)$$

we can find the individual Fourier coefficients to be

$$\begin{aligned} F_n &= \frac{1}{T}\int_{-T/2}^{T/2} \delta(t) e^{-in\omega_0 t} dt \\ &= \frac{1}{T} \end{aligned}$$

where $\omega_0 = \frac{2\pi}{T}$. Using our previous result, we can see that the Fourier transform of the Shah function is also a Shah function

$$\mathscr{F}\{f(t)\} = \omega_0 \sum_{n=-\infty}^{\infty} \delta(\omega - n\omega_0)$$

The other way to window a function is to use the windowing function $W(x)$ which is given by

$$W(x) = \begin{cases} 0 & |x| > \frac{1}{2} \\ \frac{1}{2} & |x| = \frac{1}{2} \\ 1 & |x| < \frac{1}{2} \end{cases}$$

We can then use the bijective shift

$$t = \frac{x - \frac{1}{2}(b+a)}{b-a}$$

to make the window fit to whatever sized interval we need. The Fourier transform then becomes

$$F_W(\omega) = \int_{-\infty}^{\infty} W\left(\frac{x - \frac{1}{2}(b+a)}{b-a}\right) f(x) e^{-i\omega x} dx = \int_a^b f(x) e^{-i\omega x} dx$$

and to invert, we use

$$\frac{1}{2\pi}\int_{-\infty}^{\infty} F_W(\omega) e^{i\omega x} d\omega$$

There is really no benefit to this, it just depends if you'd rather work with the Fourier series directly or with $F_W(\omega)$.

We can also define something known as the *total energy* which is defined as

$$E = \int_{-\infty}^{\infty} |f(t)|^2 dt$$

and the quantity $|F(\omega)|^2$ is referred to as the *energy spectral density* and plotting it versus ω is known as the *energy spectrum*. We can relate the energy densities between the function and its transform via Parseval's theorem for transforms (it is also called Rayleigh's theorem or Plancherel's identity).

Theorem 6.5 (Parseval's Theorem) *If $f(t)$ has a Fourier transform $F(\omega)$ and is square integrable, then*

$$\int_{-\infty}^{\infty} |f(t)|^2 dt = \frac{1}{2\pi}\int_{-\infty}^{\infty} |F(\omega)|^2 d\omega$$

Proof.

$$\begin{aligned} \int_{-\infty}^{\infty} f(t)\overline{f(t)} dt &= \int_{-\infty}^{\infty} f(t) \frac{1}{2\pi}\int_{-\infty}^{\infty} F(\omega) e^{i\omega t} dt \\ &= \frac{1}{2\pi}\int_{-\infty}^{\infty} F(\omega) \int_{-\infty}^{\infty} f(t) e^{i\omega t} dt \\ &= \frac{1}{2\pi}\int_{-\infty}^{\infty} F(\omega)\overline{F(\omega)} d\omega \\ &= \frac{1}{2\pi}\int_{-\infty}^{\infty} |F(\omega)^2| d\omega \end{aligned}$$

□

Chapter 7

Wavelets and Signal Processing

7.1 Wavelets

Another generalization of the principles of Fourier analysis, that is, decomposing a function onto a set of orthonormal basis functions, is via wavelet analysis. It was originally introduced by Alfréd Haar in 1910 by defining the Father wavelet

$$\phi(t) = \chi_{[0,1)}(t) = \begin{cases} 1 & 0 \leq t < 1 \\ 0 & \text{otherwise} \end{cases}$$

The mother wavelet is defined as

$$\psi(t) = \phi(2t) - \phi(2t-1) = \begin{cases} 1 & 0 \leq t < \frac{1}{2} \\ -1 & \frac{1}{2} \leq t < 1 \\ 0 & \text{otherwise} \end{cases}$$

The Daughter wavelets (I genuinely don't know who is the god awful person who named ts but ts genuine so cornball) and succeeding generations (again, cornball) of wavelets all look like the mother wavelet in the fact they "oscillate" (and hence the name wave-let) and not the father which looks more like a Top Hat function. There are also son wavelets which look more like the father wavelet but we will not go into too much detail about them. In general, we define the kth wavelet in the jth generation of daughter wavelets through

$$\psi_{j,k}(t) = 2^{j/2}\psi(2^j t - k)$$

where $0 \leq k \leq 2^j - 1$ and has a value of 1 in the interval $[k2^{-j}, (k+\frac{1}{2})2^{-j}]$ and -1 in the interval $[(k+\frac{1}{2})2^{-j}, (k+1)2^{-j}]$. One can verify that these are orthogonal to each other and hence linearly independent of each other.

The essence of wavelet analysis comes from representation theory (which I have zero notions about other than representing shii with linear algebra). As such, we can effectively split the interval $[0,1]$ into $2^n \ \ \forall n \in \mathbb{N}$ amount of intervals. For just a demonstration, let us split it into just four intervals. We can represent the value of a function on the interval $[0,1]$ using a 4-vector $\begin{bmatrix} a, b, c, d \end{bmatrix}^T$. We can therefore replace the basis vectors $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3, \mathbf{e}_4\}$ with a different set of basis vectors $\{\phi(t), \psi(t), \psi_{1,0}(t), \psi_{1,1}(t)\}$ which can be represented as

$$\left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ -1 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \\ -1 \end{bmatrix} \right\}$$

which are linearly independent. Since we have a collection of 4 linearly independent vectors, we know they span all of $\mathbb{R}^4$ and therefore form a basis. Hence, we can project our function onto any number of wavelets that we want.

$$f(t) = \sum_{j=-\infty}^{\infty} \sum_{k=-\infty}^{\infty} c_{jk}\psi_{j,k}$$

where the coefficient is found through the inner product between $f(t)$ and the jk^{th} wavelet

$$c_{jk} = \langle f, \psi_{j,k} \rangle = \int_{-\infty}^{\infty} f(t)\psi_{j,k}(t)dt$$

7.2 Signal Processing

We can define the auto-covariance, which measures how well a signal matches with itself at different points in time, as

$$R_f(\tau) = \int_{-\infty}^{\infty} f(t)f(t-\tau)dt$$

We also define the auto-correlation as the normalized version of the auto-covariance. That is,

$$\rho_f(\tau) = \frac{R_f(\tau)}{|R_f(\tau)|}$$

We can also trivially show using the substitution $u = t+\tau \to du = dt$ to show that the auto-covariance is a symmetric operation

$$\begin{aligned} R_f(-\tau) &= \int_{-\infty}^{\infty} f(t)f(t+\tau)dt \\ &= \int_{-\infty}^{\infty} f(u-\tau)f(u)du \\ &= R_f(\tau) \end{aligned}$$

which logically makes sense. We also define the *power spectral density* as the Fourier transform of the auto-covariance

$$S_f(\omega) = \int_{-\infty}^{\infty} R_f(\tau)e^{-i\omega\tau}d\tau$$

Suppose now that we have a 2π-periodic function such that $f_p(t) = \sum_{n=-\infty}^{\infty} f(t+2\pi n)$. We can therefore express it as a Fourier series where each coefficient is given by

$$\begin{aligned} c_k &= \frac{1}{2\pi}\int_0^{2\pi} f_p(t)e^{-ikt}dt \\ &= \frac{1}{2\pi}\int_0^{2\pi} \sum_{n=-\infty}^{\infty} f(t+2\pi n)e^{-ikt}dt \\ &= \frac{1}{2\pi}\sum_{n=-\infty}^{\infty}\int_0^{2\pi} f(t+2\pi n)e^{-ikt}dt \end{aligned}$$

assuming we impose the necessary conditions that allows us to swap the sum and integral. We now make the substitution $\xi = t + 2\pi n \to d\xi = dt$.

$$\begin{aligned} c_k &= \frac{1}{2\pi}\sum_{n=-\infty}^{\infty}\int_{2\pi n}^{2\pi(n+1)} f(\xi)e^{-ik(\xi-2\pi n)}d\xi \\ &= \frac{1}{2\pi}\int_{-\infty}^{\infty} f(\xi)e^{-ik\xi}d\xi \\ &= \frac{1}{2\pi}\hat{f}(k) \end{aligned}$$

If we generalize this to a function with any period, we arrive at Poisson's sum formula

$$\sum_{n=-\infty}^{\infty} f(t+2\pi an) = \frac{1}{2\pi a}\sum_{k=-\infty}^{\infty}\hat{f}\left(\frac{k}{a}\right)e^{ikt/a}$$

where we have replaced the n with x and then transformed with respect to x and applied Theorem 6.4 to arrive at the right hand side. Now let us show an important property known as Fourier duality. The Fourier transform of the function $\hat{f}(\omega)$ is

$$\begin{aligned} \mathscr{F}\left\{\mathscr{F}\left\{f(x)\right\}\right\} &= \int_{-\infty}^{\infty}\hat{f}(\omega)e^{-i\omega x}d\omega \\ \mathscr{F}^{-1}\left\{\hat{f}(\omega)\right\} &= \frac{1}{2\pi}\int_{-\infty}^{\infty}\hat{f}(\omega)e^{i\omega x}d\omega \\ \Rightarrow \mathscr{F}\left\{\mathscr{F}\left\{f(x)\right\}\right\} &= 2\pi f(-x) \end{aligned}$$

If we replace $f(x)$ with its transform $\hat{f}(\omega)$ in Poisson's Summation Formula and apply Fourier duality identity, we get

$$\sum_{n=-\infty}^{\infty} \hat{f}(t+2\pi an) = \frac{1}{2\pi a} \sum_{k=-\infty}^{\infty} 2\pi f\left(\frac{k}{a}\right) e^{ikt/a}$$

If we let $a = 1$ and replace t with ω and n with $-k$ (adding the minus sign doesn't matter since we are summing from $-\infty \to +\infty$), we get

$$\sum_{k=-\infty}^{\infty} \hat{f}(\omega+2\pi k) = \sum_{k=-\infty}^{\infty} f\,(k)\, e^{-ik\omega}$$

If instead we make the substitution $t = \frac{\omega}{c}$ and $a = \frac{1}{c}$, we get

$$\sum_{k=-\infty}^{\infty} \hat{f}\left(\frac{\omega+2\pi k}{c}\right) = c \sum_{k=-\infty}^{\infty} f(ck) e^{-ik\omega} \tag{7.1}$$

If instead of all of that we just plug in $a = \frac{1}{2\pi}$

$$\sum_{n=-\infty}^{\infty} f(t+n) = \sum_{k=-\infty}^{\infty} \hat{f}(2\pi k) e^{2\pi ikt}$$

Now let us consider the Fourier transform of a function such that

$$\hat{f}(2\pi k) = \delta_{0,k} \qquad k \in \mathbb{Z}$$

We can therefore see this leads to something known as the Partition of Unity

$$\sum_{n=-\infty}^{\infty} f(t+n) = 1$$

One such example is the Haar function which we have defined previously. We shall refer to it as $B_1(t) = \chi_{[0,1)}(t)$. The Fourier transform of the Haar function is

$$\hat{B}_1(\omega) = \frac{1 - e^{-i\omega}}{i\omega}$$

We can see that for all integers that aren't 0, the Fourier transform of B_1 yields 0 but $\hat{B}_1(0) = 1$. Therefore we ccan say

$$\sum_{n=-\infty}^{\infty} B_1(t+n) = 1$$

This is also known as the first order B-spline function. To find the nth order B-spline function, we use the formula

$$B_n(t) = B_{n-1}(t) * B_1(t)$$

and by using the convolution theorem for Fourier transforms and induction, the Fourier transform of the nth order B-spline function is

$$\hat{B}_n(\omega) = \left(\frac{1 - e^{-i\omega}}{i\omega}\right)^n$$

Suppose we have a signal that we wish to sample with some range of frequencies (called the *bandwidth*) 2Ω. If $h \equiv \frac{\pi}{\Omega}$, then this is the largest value h can obtain while still being able to recover the important features of a signal during reconstruction. This is due to the Shannon-Nyquist theorem and the highest frequency is called the Nyquist frequency. Let us take the Fourier transform of a signal $f(t)$ to be

$$\hat{f}(\omega) = \int_{-\infty}^{\infty} f(t) e^{-i\omega t} dt$$

We can approximate it using Simpson's rule. Let us call this approximation $\hat{F}(\omega)$

$$\hat{F}(\omega) = h \sum_{k=-\infty}^{\infty} f(kh) e^{-ik\omega h} \approx \hat{f}(\omega)$$

If we use (7.1) and let $c = h$ and $t = \omega h$ (unfortunately these equations lack some clarity, but this t gets replaced by ω when generating (7.1)), we see that

$$\hat{F}(\omega) = \sum_{k=-\infty}^{\infty} \hat{f}\left(\frac{\omega h + 2\pi k}{h}\right) = \hat{f}(\omega) + \sum_{k=-\infty}^{-1} \hat{f}\left(\omega + \frac{2\pi k}{h}\right) + \sum_{k=1}^{\infty} \hat{f}\left(\omega + \frac{2\pi k}{h}\right)$$

What we see here is that the approximation yields both the original frequency distribution $\hat{f}(\omega)$ and infinite copies spaced out by $\frac{2\pi}{h}$ on the frequency axis. To then retrieve $\hat{f}(\omega)$, we need to use a window function to isolate what we want. We let our window function in the frequency domain to be defined as

$$\hat{W}(\omega) = \begin{cases} 1 & |\omega| < \Omega \\ 0 & \text{otherwise} \end{cases}$$

We can now see why want to sample at a bandwidth of 2Ω since we want to sample fast enough such that the spacing $\frac{2\pi}{h} = 2\Omega$ allows for the copies to not overlap each other This overlapping is known as *aliasing* and is the result of undersampling. As such, we now multiply our approximation by our window function and invert to get via the convolution theorem

$$f(t) = F(t) * W(t)$$

The inverse Fourier transform of the window function is

$$W(t) = \frac{\sin(\Omega t)}{\pi t}$$

If we invert the Simpson's rule approximation, we get

$$F(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \hat{F}(\omega) e^{i\omega t} d\omega = \frac{h}{2\pi} \int_{-\infty}^{\infty} \sum_{k=-\infty}^{\infty} f(kh) e^{-ikh\omega} e^{i\omega t} d\omega$$

Assuming the conditions of being able to swap the summation and integration as well as using the identity

$$\delta(t - kh) = 2\pi \int_{-\infty}^{\infty} e^{i\omega(t-kh)} d\omega$$

We get that

$$\begin{aligned} F(t) &= \frac{h}{2\pi} \sum_{k=-\infty}^{\infty} f(kh) \int_{-\infty}^{\infty} e^{i\omega(t-kh)} d\omega \\ &= h \sum_{k=-\infty}^{\infty} f(kh) \delta(t - kh) \end{aligned}$$

Therefore our convolution becomes

$$\begin{aligned} f(t) &= h \sum_{k=-\infty}^{\infty} f(kh) \int_{-\infty}^{\infty} \delta(\tau - kh) W(t - \tau) d\tau \\ &= h \sum_{k=-\infty}^{\infty} f(kh) W(t - kh) \\ &= h \sum_{k=-\infty}^{\infty} f(kh) \frac{\sin[\Omega(t - kh)]}{\pi(t - kh)} \\ &= \sum_{k=-\infty}^{\infty} f(kh) \frac{\sin(\Omega t - k\pi)}{\Omega t - k\pi} \end{aligned}$$

We shall now investigate the Gibbs phenomenon which is the overshoot at the discontinuities of a truncated Fourier series.

$$f_N(t) = \sum_{k=-N}^{N} c_k e^{ik\omega_0 t}$$

The coefficients are found in the usual manner and if we substitute everything in and swap the order of summation and integration, we arrive at

$$\begin{aligned} f_N(t) &= \sum_{k=-N}^{N} \frac{1}{T} \int_{-T/2}^{T/2} f(\tau) e^{-ik\omega_0 \tau} e^{ik\omega_0 t} d\tau \\ &= \frac{1}{T} \int_{-T/2}^{T/2} f(\tau) \sum_{k=-N}^{N} e^{ik\omega_0 (t-\tau)} d\tau \\ &= \frac{1}{T} \int_{-T/2}^{T/2} f(\tau) \frac{\sin\left[(N+\frac{1}{2})(t-\tau)\omega_0\right]}{\sin\left[\frac{1}{2}(t-\tau)\omega_0\right]} \end{aligned}$$

We get the fraction with sin functions through recognizing the sum as a geometric series. It is also the this fraction of sines that leads to the Gibbs phenomenon noted at the discontinuities.

7.3 Localization

We can define the *center* of a distribution as

$$\mu_p = \frac{1}{||p(t)||^2} \int_{-\infty}^{\infty} t|p(t)|^2 dt$$

where $||p(t)||^2$ is the norm of $p(t)$ defined as

$$||p(t)||^2 = \int_{-\infty}^{\infty} |p(t)|^2 dt$$

We can also define the root-mean-square (RMS)

$$\Delta_p = \frac{1}{||p(t)||} \left[\int_{-\infty}^{\infty} (t-\mu)^2 |p(t)|^2 dt \right]^{1/2}$$

which acts as the spread of frequency distribution. This leads the uncertainty principle where the product of the RMS of a function and its Fourier transform is greater than half

$$\Delta_f \Delta_{\hat{f}} \geq \frac{1}{2}$$

In the case that one of the RMS is equal to zero, the other must be infinite.

7.4 Short Time Fourier Transform

To start, let us take a function $f(t)$ and multiply it by a window function $b(t)$. Let us call

$$f_b(t) = f(t)b(t)$$

and

$$b_{t_0,\omega}(t) = b(t-t_0)e^{i\omega t}$$

We can define the Short Time Fourier Transform (STFT) as

$$f_G(t_0,\omega) = \int_{-\infty}^{\infty} f(t)b(t-t_0)e^{-i\omega t} dt = \int_{-\infty}^{\infty} f(t)\overline{b_{t_0,\omega}(t)} dt$$

The G represents the relation with the Green's Functions. That is because the regular Fourier transform makes the assumption that a signal is static and time invariant; however, functions are rarely like this and so we want to find the Green's function of a linear time-varying system that depends on both t and the time of the impulse τ. Ideally we would be able to freeze time and figure this out to get a transfer function. That is why we use the STFT which

takes a short period of time to find the time-varying transfer function of the system. We invert to get back the windowed function very easily using

$$f_b(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} f_G(t_0,\omega)e^{i\omega t}d\omega$$

If we replace t_0 with the dummy variable τ, the STFT looks like

$$f_{G_b}(\tau,\xi) = \int_{-\infty}^{\infty} f(t)\overline{b_{\tau,\xi}(t)}dt = \int_{-\infty}^{\infty} f(t)b(t-\tau)e^{-i\xi t}dt = \int_{-\infty}^{\infty} f_b(t)e^{i\xi t}dt$$

and the inversion formula being

$$f_b(\tau,t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} f_{G_b}(\tau,\xi)e^{i\xi t}d\xi$$

Let us take this inversion formula and multiply the complex conjugate of $b(t-\tau)$

$$|b(t-\tau)|^2 f(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} f_{G_b}(\tau,\xi)\overline{b(t-\tau)}e^{i\xi t}d\xi$$

We assume the window function is real valued (for simplicity's sake) so the complex conjugate is just itself. We therefore have

$$\begin{aligned}
|b(t)|^2 f(t) &= \frac{1}{2\pi}\int_{-\infty}^{\infty} f_{G_b}(\tau,\xi)b_{\tau,\xi}(t)d\xi \\
f(t)\int_{-\infty}^{\infty}|b(t-\tau)|^2 d\tau &= \frac{1}{2\pi}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f_{G_b}(\tau,\xi)b_{\tau,\xi}(t)d\xi d\tau \\
f(t) &= \frac{1}{2\pi||b(t)||^2}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f_{G_b}(\tau,\xi)b_{\tau,\xi}(t)d\xi d\tau
\end{aligned}$$

Chapter 8

Complex Analysis

8.1 Introduction

This will by no means be a formal introduction to complex analysis and I plan to study the subject further in detail at a later point (hopefully with Stein and Shakarchi Complex). A complex valued $f(z)$ where $z = x + iy$ can be written in the form

$$f(z) = \phi(x, y) + i\psi(x, y)$$

where $\phi(x, y)$ gives you the real part of the function and $\psi(x, y)$ gives you the imaginary part. Both are real valued functions. A real valued function is deemed *analytic* or *regular* if one can represent it as a power series. This means that the function is infinitely differentiable. The complex analog is the term *holomorphic*, but is effectively interchangeable with analytic.

Theorem 8.1 *A complex-valued function $f(z) = \phi(x, y) + i\psi(x, y)$ (where $z = x + iy$) is holomorphic if and only if it satisfies the Cauchy-Riemann Equations*

$$\frac{\partial \phi}{\partial x} = \frac{\partial \psi}{\partial y} \qquad \frac{\partial \phi}{\partial y} = -\frac{\partial \psi}{\partial x}$$

Proof. ($\Rightarrow$) We start with the fact that the definition of being holomorphic implies infinite differentiability. For this to happen, we require that the limit definitions of the derivatives coming along different paths to converge to the same value. Let us only consider coming along the real axis and imaginary axis first. Coming along the real axis, we have

$$f'(z_0) = \lim_{\Delta x \to 0} \frac{[\phi(x_0 + \Delta x, y_0) + i\psi(x_0 + \Delta x, y_0)] - [\phi(x_0, y_0) + i\psi(x_0, y_0)]}{\Delta x}$$

and along the imaginary

$$f'(z_0) = \lim_{\Delta y \to 0} \frac{[\phi(x_0, y_0 + \Delta y) + i\psi(x_0, y_0 + \Delta y)] - [\phi(x_0, y_0) + i\psi(x_0, y_0)]}{i\Delta y}$$

Setting these equal to each other and finishing the limit, we get

$$f'(z_0) = \frac{\partial \phi}{\partial x} + i\frac{\partial \psi}{\partial x} = \frac{\partial \psi}{\partial y} - i\frac{\partial \phi}{\partial y}$$

Setting the real and imaginary parts equal to each other, we get

$$\frac{\partial \phi}{\partial x} = \frac{\partial \psi}{\partial y} \qquad \frac{\partial \phi}{\partial y} = -\frac{\partial \psi}{\partial x}$$

($\Leftarrow$) Assume $f(z)$ has derivatives $\{\phi_x, \phi_y, \psi_x, \psi_y\}$ that are continuous. We can approximate a change in ϕ and ψ using the following

$$\Delta\phi = \phi_x \Delta x + \phi_y \Delta y + \varepsilon_1 \Delta z$$
$$\Delta\psi = \psi_x \Delta x + \psi_y \Delta y + \varepsilon_2 \Delta z$$

where $\varepsilon_1, \varepsilon_2$ are error terms that go to zero as $\Delta z = \Delta x + i\Delta y$ goes to zero. We can represent the change in $f(z)$ as

$$\begin{aligned}\Delta f = \Delta\phi + i\Delta\psi &= (\phi_x \Delta x + \phi_y \Delta y + \varepsilon_1 \Delta z) + i(\psi_x \Delta x + \psi_y \Delta y + \varepsilon_2 \Delta z) \\ &= (\phi_x + i\psi_x)\Delta x + (\phi_y + i\psi_y)\Delta y + (\varepsilon_1 + i\varepsilon_2)\Delta z\end{aligned}$$

Now subbing in the relations $\phi_x = \psi_y$ and $\phi_y = -\psi_x$ for the Δy terms and subbing $\eta = \varepsilon_1 + i\varepsilon_2$

$$\Delta f = (\phi_x + i\psi_x)\Delta x + (-\psi_x + i\phi_x)\Delta y + \eta \Delta z$$

If we factor an i out of the Δy term, we get

$$\begin{aligned}\Delta f &= (\phi_x + i\psi_x)\Delta x + i(i\psi_x + \phi_x)\Delta y + \eta \Delta z \\ &= (\phi_x + i\psi_x)(\Delta x + \Delta y) + \eta \Delta z \\ &= [(\phi_x + i\psi_x) + \eta]\Delta z \\ \Rightarrow \frac{\Delta f}{\Delta z} &= \phi_x + i\psi_x + \eta\end{aligned}$$

Taking the limit, we force the error terms to zero and we get

$$\lim_{\Delta z \to 0} \frac{\Delta f}{\Delta z} = f'(z) = \phi_x + i\psi_x$$

Since this limit exists and converges to the same value regardless of what path taken, the function is analytic and therefore holomorphic. □

Let us now consider a complex-valued function that is analytic in an annular region $r_1 < |z - z_0| < r_2$. This series possesses a *Laurent Series*

$$f(z) = \sum_{n=-\infty}^{\infty} a_n (z - z_0)^n$$

The positive n parts are deemed the Taylor part of the Laurent series and the terms associated with negative are deemed the principal part. The coefficient a_{-1} is termed the *residue* and will be of great importance as shown later. As said previously, holomorphic functions are harmonic. These follow from the Cauchy-Riemann equations as follows

$$\begin{aligned}\phi_x = \psi_y \qquad \phi_y = -\psi_x \\ \phi_{xx} = \psi_{xy} \qquad \phi_{yy} = -\psi_{yx}\end{aligned}$$

And assuming the partial derivatives are all continuous, we apply Clairaut's theorem to retrieve that

$$\begin{aligned}\phi_{xx} = \psi_{xy} &= -\phi_{yy} \\ \Rightarrow \nabla^2 \phi &= 0\end{aligned}$$

Proving that the imaginary part ψ is harmonic is done via a similar method. Since they are both harmonic functions, we can express the value of $f(z)$ at a point $z = z_0$ using the values of $f(z)$ on some boundary C by the use of Cauchy's integral formula

$$f^{(n)}(z_0) = \frac{n!}{2\pi i} \int_C \frac{f(z)}{(z - z_0)^{n+1}} dz$$

Note that $f(z)$ in this case must be analytic everywhere inside of C. If we consider a function $g(z)$ such that is not analytic at $z = z_0$

$$g(z) = \frac{f(z)}{(z - z_0)^n}$$

we say that there is a singularity at $z = z_0$. We say that the singularity is a *pole of order n*. If $n = \infty$, then we call that an *essential* singularity and we cannot use the Cauchy integral formula.

8.2 Complex Integration

Suppose $f(t)$ is a complex valued function of the real variable $t \in [a,b]$ with real part f_r and imaginary part p_i which are both real valued, piecewise continuous functions on $[a,b]$. We can integrate $f(t)$ using the line integrals

$$\int_a^b f(t)dt = \int_a^b f_r(t)dt + i\int_a^b f_i(t)dt \tag{8.1}$$

Let us now walk through some important definitions.

Definition 8.2

1. An arc C is a set of points $\{(x(t),y(t)) : t \in [a,b]\}$ where $x(t)$ and $y(t)$ are continuous functions of the real variable t. The arc is conveniently described in terms of the complex valued function z of the real variable t
2. An arc is *simple* if it does not cross itself. That is $z(t_1) = z(t_2) \Rightarrow t_1 = t_2 \ \forall t_1, t_2 \in [a,b]$.
3. An arc is *smooth* if $z'(t)$ exists and is non-zero for $t \in [a,b]$. This ensures that C has a continuously turning tangent.
4. A (simple) contour is an arc consisting of a finite number of (simple) smooth arcs joined end to end. When only the initial and final values of $z(t)$ coincide, the contour is a (simple) closed contour.

Definition 8.3 Let C be a simple contour as defined above extending from the point $\alpha = z(a)$ to $\beta = z(b)$. Let a domain $\mathcal{D}$ be a subset of the complex plane and let the curve C lie wholly within it. Define $f(z)$ to be a piecewise continuous function on C, that is $f(z(t))$ is piecewise continuous on the interval $[a,b]$. Then the *contour integral* of $f(z)$ is

$$\int_C f(z)dz = \int_a^b f(z)dz = \int_a^b f(z(t))z'(t)dt$$

We shall now prove Cauchy's theorem which is a vital result from complex analysis.

Theorem 8.4 *If $f(z)$ is analytic in a domain D and on its closed boundary C, then*

$$\oint_C f(z)dz = 0$$

Cauchy's Theorem. There a variety of different proofs but the most straightforward uses Green's theorem. Green's theorem states that if $P(x,y)$ and $Q(x,y)$ are C^1 functions in D and on C, then

$$\oint_C Pdx + Qdy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dxdy$$

Let us consider $f(z) = P(x,y) + iQ(x,y)$ where $z = x + iy$. We then have

$$\begin{aligned} f(z)dz &= (P(x,y) + iQ(x,y))(dx + idy) \\ &= P(x,y)dx - Q(x,y)dy + i(P(x,y)dy + Q(x,y)dx) \\ \Rightarrow \oint_C f(z)dz &= \oint_C (P(x,y) - Q(x,y)dy) + i\oint_C (P(x,y)dy + Q(x,y)dx) \\ &= \iint_D \left(-\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dxdy + i\iint_D \left(\frac{\partial P}{\partial x} - \frac{\partial Q}{\partial y}\right) dxdy \end{aligned}$$

The Cauchy-Riemann equations imply $P_x = Q_y$ and $P_y = -Q_x$ which then gives

$$\oint_C f(z)dz = 0$$

□

Using this, we shall now prove the most important result from complex analysis.

Theorem 8.5 (Residue Theorem) *If $f(z)$ is analytic within and on C except at points z_i such that $i \in \{1, \ldots, n\}$ lie inside C where $f(z)$ has poles, then*

$$\oint_C f(z)dz = 2\pi i \sum_{k=1}^{n} (\textit{sum of residues of } f(z) \textit{ at } z = z_k)$$

Proof. Let us take a contour C that encompasses a region of space in the complex plane with just one pole (as this method generalizes to finite number of poles). Let us now consider the contour C' that is mostly the same as C but has protrusion such that it does not actually contain the pole located at $z = z_0$. We can therefore say that $f(z)$ is

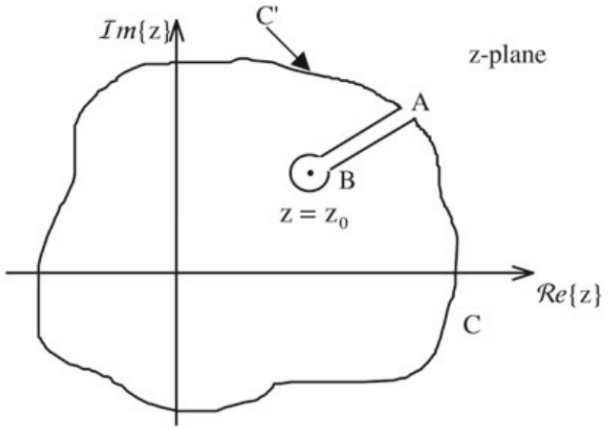


Figure 8.1: Contour that goes around the pole

analytic everywhere in C'. As such, we can apply Cauchy's theorem. We can also break up our contour integral into smaller line integrals

$$\oint_{C'} = \int_C + \int_{AB} + \int_{BA} - \int_\gamma = 0$$

The negative sign comes from the fact that we are going clockwise around the pole. Since the AB and BA integrals are of the same magnitude and go in opposite direction, their sum is equal to zero and thus we have

$$\int_C f(z)dz = \int_\gamma f(z)dz$$

where γ is the small circle of radius ϵ surrounding $z = z_0$. We can represent $f(z)$ through its Laurent series and evaluate the integral term by term (which is valid since the Laurent series is uniformly convergent). We therefore evaluate each integral as

$$\int_\gamma (z - z_0)^k dz \qquad k \in \mathbb{Z}$$

On the path γ, $z - z_0 = \epsilon e^{i\theta}$ where $\theta \in [0, 2\pi)$ and ϵ is the radius of path. We therefore have that $dz = i\epsilon e^{i\theta} d\theta$

$$\begin{aligned}\int_\gamma (z - z_0)^k dz &= \int_0^{2\pi} i e^{k+1} e^{(k+1)i\theta} d\theta \\ &= \frac{i\epsilon^{k+1}}{i(k+1)} e^{i(k+1)\theta}\Big|_0^{2\pi}\end{aligned}$$

If $k \neq -1$, the integral evaluates to 0. If $k = 1$, we have something different

$$\int_\gamma (z - z_0)^{-1} dz = \int_0^{2\pi} \frac{i\epsilon e^{i\theta}}{\epsilon e^{i\theta}} d\theta = 2\pi i$$

and so therefore the integral

$$\int_\gamma f(z)dz = 2\pi i a_{-1}$$

Now if there are finite amount of poles, we do this same procedure for each one and so thus we achieve the desired result

$$\int_C f(z)dz = 2\pi i \sum_{k=1}^{n} \text{Res}(f, z_k)$$

□

8.3 Branch Cuts

Due to the nature of complex numbers, certain functions may end up being multi-valued. Functions like this include the natural logarithm and root functions. One can search up images of these functions by looking up "Riemann Sheets of Complex Functions." What generally happens is we define the negative real axis to be where we place this branch cut and have $\theta = -\pi$ be defined for some value and $\theta = \pi$ be defined for another value. Let us also take into account how we take the natural log of a complex number. Recall that any complex number z can be represented as $z = re^{i\theta}$

$$\begin{aligned} \ln(z) &= \ln(re^{i\theta}) \\ &= \ln(r) + i\theta \\ &= \ln(r) + i(\vartheta + 2\pi k) \qquad k \in \mathbb{Z} \end{aligned}$$

8.4 Inverse Laplace Transform

Theorem 8.6 *If the Laplace transform of $f(t)$ exists, then*

$$f(x) = \lim_{k\to\infty} \left\{ \frac{1}{2\pi i} \int_{\sigma-ik}^{\sigma+ik} \bar{f}(s)e^{st}ds \right\} \quad t > 0$$

where $|f(x)| \leq e^{Mx}$ for some positive real number M and σ is another real number such that $\sigma > M$.

Proof. The proof was already given previously but we shall restate it.

Let us consider a complex valued function $F(\omega)$ of the form

$$F(\omega) = A(\omega)e^{i\phi(\omega)}$$

where $A(\omega)$ and $\phi(\omega)$ are real value functions of $\omega \in \mathbb{R}$. $A(\omega)$ acts as the amplitude and $\phi(\omega)$ acts as the phase angle. Using the inverse Fourier transform, we see that

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} A(\omega)e^{i\omega x + i\phi(\omega)}d\omega$$

However, this is often not very useful since the oscillatory nature leads to a lot of integrals not converging. As such, let instead do the Fourier transform of the function $e^{-kx}f(x)$ where the exponential decay term allows more functions to converge. We then change the lower bound of integration to 0 and assuming $f(x)$ is of exponential order, we have

$$F_k(\omega) = \int_0^{\infty} e^{-(k+i\omega)x} f(x)dx$$

Finding the inverse of F_k using our definition for the inverse Fourier transform gives us

$$\frac{1}{2\pi} \int_{-\infty}^{\infty} e^{i\omega x} F_k(\omega)d\omega = \begin{cases} 0 & x < 0 \\ e^{-kx}f(x) & x \geq 0 \end{cases}$$

Multiplying both sides to cancel out the damping factor on the right hand side (and bringing it under the integral side on the left), we have

$$\frac{1}{2\pi} \int_{-\infty}^{\infty} e^{(k+i\omega)x} F_k(\omega)d\omega = \begin{cases} 0 & x < 0 \\ f(x) & x \geq 0 \end{cases}$$

We now make the substitution $s = k + i\omega \to ds = id\omega$ so now the limits of integration will be $s = k \pm i\infty$ when $\omega = \pm\infty$ where we now have

$$f(x) = \frac{1}{2\pi i}\int_{k-i\infty}^{k+i\infty} e^{sx}F(s)ds$$

□

To use this theorem, we use the Bromwich contour which is a circular contour with a vertical path occurring at the real number σ. To find the inverse Laplace transform of a function, we simply evaluate the integral along the Bromwich contour (using Cauchy's or Residue theorem) before breaking up the integral and solving for the inversion integral shown above.

8.5 The Inversion Formula in Asymptotics

Theorem 8.7 *If $\bar{f}(s)$ is $O\left(\frac{1}{|s|}\right)$ as $s \to \infty$, the singularity with the largest real part is at $s = s_1$, and in some neighborhood of s_1*

$$\bar{f}(s) = (s - s_1)^k \sum_{n=0}^{\infty} a_n (s - s_1)^n \qquad -1 < k < 0$$

then

$$f(t) = -\frac{1}{\pi} e^{s_1 t} \sin(k\pi) \sum_{n=0}^{\infty} a_n (-1)^n \frac{\Gamma(n + k + 1)}{t^{n+k+1}} \text{ as } t \to \infty$$

Proof. We wish to evaluate the integral

$$\frac{1}{2\pi i}\int_C e^{st}\bar{f}(s)ds$$

as the large circular arc of the Bromwich contour goes to zero by Jordan's Lemma and through other manipulations, we can find the inverse Laplace transform formula. Since $0 < k < 1$, the function is multi-valued around s_1 and so thus we must introduce a branch cut such that on the lower Riemann sheet we have $s = s_1 + xe^{-i\pi}$ and on the upper sheet we have $s = s_1 + xe^{i\pi}$. Let us now evaluate the "indentation" integrals along each Riemann sheet

$$f(t) = \frac{1}{2\pi i}\int_{\infty}^{0} e^{(s_1 - x)t}\bar{f}(s_1 + xe^{-i\pi})e^{-i\pi}dx + \frac{1}{2\pi i}\int_{0}^{\infty} e^{(s_1 - x)t}\bar{f}(s_1 + xe^{i\pi})e^{i\pi}dx$$

where the $e^{\pm i\pi}$ next to the differentials come from the substitution $s = s_1 + xe^{\pm i\pi}$. We now combine the integrals into one using properties of definite integrals to get

$$f(t) = -\frac{1}{2\pi i}\int_{0}^{\infty} e^{(s_1 - x)t}[\bar{f}(s_1 + xe^{i\pi}) - \bar{f}(s_1 + x^{-i\pi})]dx$$

We now use Watson's Lemma and find the value of $f(s)$ around $s = s_1$

$$\bar{f}(s_1 + xe^{i\pi}) = (xe^{i\pi})^k \sum_{n=0}^{\infty} a_n (xe^{i\pi})^n$$

$$\bar{f}(s_1 + xe^{-i\pi}) = (xe^{-i\pi})^k \sum_{n=0}^{\infty} a_n (xe^{-i\pi})^n$$

$$\Rightarrow \bar{f}(s_1 + xe^{i\pi}) - \bar{f}(s_1 + xe^{-i\pi}) = x^k 2i\sin(k\pi)\sum_{n=0}^{\infty} a_n (-1)^n x^n$$

where we note that

$$\sin(x) = \frac{1}{2i}(e^{ix} - e^{-ix})$$

Performing the actual integration, we get

$$\begin{aligned}\frac{1}{2\pi i}\int_0^\infty e^{(s_1-x)t}[\bar{f}(s_1+xe^{i\pi})-\bar{f}(s_1+x^{-i\pi})]dx &= \frac{\sin(k\pi)}{\pi}\int_0^\infty \sum_{n=0}^\infty a_n(-1)^n x^{n+k}e^{(s_1-x)t}dx \\ &= \frac{\sin(k\pi)}{\pi}e^{s_1 t}\sum_{n=0}^\infty a_n(-1)^n\int_0^\infty x^{n+k}e^{-xt} \\ &= \frac{\sin(k\pi)}{\pi}e^{s_1 t}\sum_{n=0}^\infty a_n(-1)^n\frac{\Gamma(n+k+1)}{t^{n+k+1}}\end{aligned}$$

where we have assumed the conditions of swapping the summation and integration operators and used the definition of the Gamma function

$$\Gamma(n)=\int_0^\infty u^{n-1}e^{-u}du$$

where in the integral in the problem, we made the substitution $u=xt\rightarrow du=tdx$ such that

$$\begin{aligned}\int_0^\infty x^{n+k}e^{-xt}dx &= \int_0^\infty \left(\frac{u}{t}\right)^{n+k}e^{-u}\frac{du}{t} \\ &= \frac{1}{t^{n+k+1}}\int_0^\infty u^{n+k}e^{-u}du \\ &= \frac{\Gamma(n+k+1)}{t^{n+k+1}}\end{aligned}$$

Putting everything together, we have

$$f(t)=-\frac{\sin(k\pi)}{\pi}e^{s_1 t}\sum_{n=0}^\infty a_n(-1)^n\frac{\Gamma(n+k+1)}{t^{n+k+1}}$$

□